

T(R)IENERMENTS: TERNARY GENERALIZATIONS OF TOURNAMENTS AND THEIR ISOMORPHISM THEORY

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ABSTRACT. A *t(r)ienement* is a complete graph on n vertices in which each edge is assigned one of three states: a directed arc $u \rightarrow v$, a directed arc $v \rightarrow u$, or a bidirectional *tie* $u \leftrightarrow v$. Ties allow Hamiltonian paths to be formed through them in either direction, making t(r)ienements a natural ternary generalization of tournaments.

We derive a closed Burnside formula for the number of unlabeled isomorphism classes, which equals the sequence [A007025](#) of unlabeled oriented graphs. We establish a canonical isomorphism between “positive” t(r)ienements (ties as $\leftrightarrow$) and “negative” t(r)ienements (ties as absent edges), explaining why both spaces are counted by the same sequence.

We introduce and compute the refined distribution $f(n, k)$, the number of isomorphism classes on n vertices with exactly k ties. The $k = 0$ slice recovers the tournament count [A000568](#), while $f(n, \binom{n}{2}) = f(n, \binom{n}{2} - 1) = 1$ for all n . The tail values stabilize: $f(n, \binom{n}{2} - 2) = 4$ for $n \geq 4$, and $f(n, \binom{n}{2} - 3) = 14$ for $n \geq 6$.

We prove that every t(r)ienement has at least one directed Hamiltonian path (extending Rédei’s theorem), and show that, unlike tournaments, the path count $H(\mathcal{T})$ need not be odd. The Grinberg–Stanley Odd-Cycle Collection Formula $H(\mathcal{T}) = I(\Omega(D_{\mathcal{T}}), 2)$ extends to all t(r)ienements via the associated digraph $D_{\mathcal{T}}$.

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1. INTRODUCTION

A *tournament* on n vertices is a complete directed graph: for every pair $\{u, v\}$, exactly one of $u \rightarrow v$ or $v \rightarrow u$ holds. Tournaments model ranking situations (round-robin competitions, paired comparisons, elections) in which every match produces a strict winner.

Real-world ranking data, however, frequently involves *ties*: two alternatives that cannot be separated by the comparison mechanism. A referee may declare a draw; two items may score identically on a metric; rounding in a sports table may leave two teams indistinguishable. The natural question is how to extend tournament theory to accommodate a third outcome.

1.1. Two representations of ties. There are two canonical ways to incorporate ties into a complete graph on n vertices.

- **Positive representation (t(r)ienerments).** Each pair $\{u, v\}$ receives one of three labels: $u \rightarrow v$, $v \rightarrow u$, or $u \leftrightarrow v$. Bidirectional edges are the “positive” encoding of a tie: they exist in both directions and may be traversed freely by a Hamiltonian path.

- **Negative representation (oriented graphs).** Each pair $\{u, v\}$ receives one of three labels: $u \rightarrow v$, $v \rightarrow u$, or $\emptyset$ (no arc). Absent edges encode ties: they represent outcomes that were never decided. These objects are precisely the *oriented graphs* (simple digraphs with no symmetric pair of arcs) studied in graph theory.

A central observation of this paper (Theorem 5.1) is that these two representations give *isomorphic* combinatorial spaces: the S_n -equivariant bijection that swaps ties and absent edges preserves all structure, so the two spaces have identical isomorphism-class counts.

1.2. The name. We call an object in the positive representation a *t(r)ienerment*, blending “tournament” with the prefix “tri” (three outcomes) and a nod to the Wiener score used in some tournament-ranking contexts. The parenthesised “r” signals both “ternary” and the alternative reading “trienerment” (three-outcome tournament).

1.3. Summary of results.

- (1) The number of unlabeled isomorphism classes of n -vertex t(r)ienerments is

$$T(n) = \frac{1}{n!} \sum_{\sigma \in S_n} 3^{B(\sigma)}, \quad B(\sigma) = \sum_a \lfloor \frac{l_a-1}{2} \rfloor + \sum_{a < b} \gcd(l_a, l_b),$$

where $(l_1, l_2, \dots)$ is the cycle type of σ . The sequence $T(n)$ is [A007025](#).

- (2) The refined distribution $f(n, k)$ (iso classes with k ties) is computed from a generating polynomial per cycle type, with $f(n, 0) = \text{A000568}(n)$ and $f(n, \binom{n}{2}) = f(n, \binom{n}{2} - 1) = 1$.
- (3) Every t(r)ienerment has at least one directed Hamiltonian path (Rédei’s theorem extends), but $H(\mathcal{T})$ need not be odd.
- (4) The Odd-Cycle Collection Formula extends: $H(\mathcal{T}) = I(\Omega(D_{\mathcal{T}}), 2)$ for the associated digraph $D_{\mathcal{T}}$.

1.4. Relation to existing work. Tournament isomorphism classes are counted by the Burnside formula

$$\text{A000568}(n) = \frac{1}{n!} \sum_{\substack{\sigma \in S_n \\ \text{all cycles odd}}} 2^{B(\sigma)},$$

where the restriction to odd-cycle permutations arises because even cycles force a symmetric pair of arcs, which is forbidden in tournaments. In t(r)ienerments, bidirectional ties resolve this obstruction, so all of S_n contributes with weight $3^{B(\sigma)}$.

The sequence $T(n) = \text{A007025}(n)$ counts unlabeled oriented graphs, which appear in the context of “semi-complete digraphs” and “tournaments with draws” in social choice theory. The connection to the triangular distribution $f(n, k)$ appears to be new.

1.5. Organization. Section 2 sets up formal definitions. Section 3 derives the Burnside formula. Section 4 develops the refined distribution $f(n, k)$. Section 5 proves the positive-negative isomorphism. Section 6 discusses Hamiltonian paths and the OCF extension. Section 7 collects the computed tables. Section 8 lists open questions.

2. DEFINITIONS

Definition 2.1 (T(r)ienerment). Let $V = \{1, \dots, n\}$. A *t(r)ienerment* on V is a function

$$\tau : \binom{V}{2} \rightarrow \{“u \rightarrow v”, “v \rightarrow u”, “u \leftrightarrow v”\}$$

that assigns to each unordered pair $\{u, v\}$ one of three states. We write:

- $u \xrightarrow{\tau} v$ if $\tau(\{u, v\}) = “u \rightarrow v”$,
- $v \xrightarrow{\tau} u$ if $\tau(\{u, v\}) = “v \rightarrow u”$,
- $u \xleftrightarrow{\tau} v$ if $\tau(\{u, v\}) = “u \leftrightarrow v”$.

The number of *ties* in τ is $k(\tau) = |\{\{u, v\} : \tau(\{u, v\}) = “u \leftrightarrow v”\}|$.

Remark 2.2. When $k(\tau) = 0$, a t(r)ienerment is exactly a tournament.

Definition 2.3 (Isomorphism). Two t(r)ienerments τ, τ' on V are *isomorphic*, written $\tau \cong \tau'$, if there exists a permutation $\sigma \in S_n$ such that for all $\{u, v\} \in \binom{V}{2}$, $\tau'(\{\sigma(u), \sigma(v)\})$ is the same state as $\tau(\{u, v\})$ with u and v replaced by $\sigma(u)$ and $\sigma(v)$. (In particular, the state “ $u \rightarrow v$ ” maps to “ $\sigma(u) \rightarrow \sigma(v)$ ”, and the state “ $u \leftrightarrow v$ ” maps to “ $\sigma(u) \leftrightarrow \sigma(v)$ ”.)

Definition 2.4 (Associated digraph). The *associated digraph* D_τ of a t(r)ienerment τ is the digraph on V with arc set

$$A(D_\tau) = \{(u, v) : u \xrightarrow{\tau} v \text{ or } u \xleftrightarrow{\tau} v\}.$$

Each directed edge $u \rightarrow v$ in τ contributes the single arc (u, v) ; each tie $u \leftrightarrow v$ contributes both arcs (u, v) and (v, u) .

Definition 2.5 (Directed Hamiltonian path). A *directed Hamiltonian path* of a t(r)ienerment τ is a sequence $v_1, v_2, \dots, v_n$ of all n vertices such that $(v_i, v_{i+1}) \in A(D_\tau)$ for $i = 1, \dots, n-1$. Equivalently, for each consecutive pair (v_i, v_{i+1}) , either $v_i \xrightarrow{\tau} v_{i+1}$ or $v_i \xleftrightarrow{\tau} v_{i+1}$. We denote by $H(\tau)$ the number of directed Hamiltonian paths in τ .

Definition 2.6 (Negative t(r)ienerment / oriented graph). A *negative t(r)ienerment* (or *oriented graph*) on V is a function

$$\nu : \binom{V}{2} \rightarrow \{“u \rightarrow v”, “v \rightarrow u”, \emptyset\}$$

where $\emptyset$ represents a *tie by absence*: no arc between u and v .

3. COUNTING ISOMORPHISM CLASSES VIA BURNSIDE

Let $\mathfrak{T}_n$ denote the set of all labeled t(r)ienerments on $V = \{1, \dots, n\}$. We have $|\mathfrak{T}_n| = 3^{\binom{n}{2}}$. The group S_n acts on $\mathfrak{T}_n$ by relabeling vertices: $(\sigma \cdot \tau)(\{u, v\}) = \tau(\{\sigma^{-1}(u), \sigma^{-1}(v)\})$, with the state's orientation updated accordingly. The number of isomorphism classes is given by Burnside's lemma:

$$T(n) = \frac{1}{n!} \sum_{\sigma \in S_n} |\text{Fix}(\sigma)|,$$

where $\text{Fix}(\sigma) = \{\tau \in \mathfrak{T}_n : \sigma \cdot \tau = \tau\}$.

3.1. Orbit analysis for a fixed permutation. Fix $\sigma \in S_n$ with cycle type $(l_1, l_2, \dots, l_m)$ (so $\sum l_a = n$). We analyze the orbits of σ acting on ordered pairs (u, v) with $u \neq v$.

Definition 3.1. For an orbit O of σ on ordered pairs, let $O^{\text{rev}} = \{(v, u) : (u, v) \in O\}$.

- O is a *Case A orbit* if $O = O^{\text{rev}}$.
- If $O \neq O^{\text{rev}}$, then (O, O^{rev}) is a *Case B orbit pair*.

The orbit analysis splits into within-cycle and cross-cycle orbits.

3.1.1. Within-cycle orbits. For a cycle $C = (c_0, c_1, \dots, c_{l-1})$ of length l , the orbit of the pair (c_0, c_k) under σ has size l and is $O_k = \{(c_i, c_{(i+k) \bmod l}) : i = 0, \dots, l-1\}$. Its reverse is O_{l-k} .

Lemma 3.2. *In a cycle of length l :*

- (1) *If l is even, then $O_{l/2}$ is the unique Case A orbit. It covers $l/2$ unordered pairs. There are $\lfloor (l-1)/2 \rfloor$ Case B orbit pairs, each covering l unordered pairs.*
- (2) *If l is odd, there are no Case A orbits. There are $(l-1)/2$ Case B orbit pairs, each covering l unordered pairs.*

In both cases, the number of Case B orbit pairs within a cycle of length l is $\lfloor (l-1)/2 \rfloor$.

3.1.2. Cross-cycle orbits. For two cycles C_a of length l_a and C_b of length l_b ($a \neq b$), the orbit of a cross-pair (c_a^i, c_b^j) has size $\text{lcm}(l_a, l_b)$.

Lemma 3.3. *Between cycles C_a and C_b , there are exactly $\text{gcd}(l_a, l_b)$ Case B orbit pairs, each covering $\text{lcm}(l_a, l_b)$ unordered pairs. There are no Case A cross-cycle orbits.*

Proof. The number of forward orbits from C_a to C_b is $l_a l_b / \text{lcm}(l_a, l_b) = \text{gcd}(l_a, l_b)$. Since $C_a \neq C_b$, no cross-orbit can equal its own reverse: (u, v) with $u \in C_a, v \in C_b$ has (v, u) in the reverse orbit which lies from C_b to C_a , a distinct family. $\square$

3.1.3. Consequences for fixed $t(r)$ ienements. A labeled $t(r)$ ienement τ is fixed by σ if and only if τ is constant on each orbit of σ on ordered pairs (respecting the directional encoding: the state “ $u \rightarrow v$ ” is sent to “ $\sigma(u) \rightarrow \sigma(v)$ ”).

Lemma 3.4. *For a Case A orbit $O = O^{\text{rev}}$ covering a set P of unordered pairs: τ is fixed by σ on P if and only if all pairs in P are ties. There is exactly one such choice.*

Proof. If $(u, v) \in O$ and $(v, u) \in O$, then fixing τ requires $\tau(\{u, v\})$ to be consistent with both orderings. The states “ $u \rightarrow v$ ” and “ $v \rightarrow u$ ” would need to simultaneously hold, which is impossible. Only the symmetric state “ $u \leftrightarrow v$ ” is consistent. Since τ is constant on O , all pairs in P are forced to be ties. $\square$

Lemma 3.5. *For a Case B orbit pair (O, O^{rev}) covering a set P of unordered pairs: τ is fixed by σ on P if and only if τ assigns the same state to all pairs in P , and that state may be any of the three options. There are exactly three choices.*

In terms of ties: either all pairs in P are directed (two choices: constant “ $u \rightarrow v$ ” direction or constant “ $v \rightarrow u$ ” direction, each self-consistent), or all pairs in P are ties.

Proof. Since O and O^{rev} are distinct orbits, the values of τ on (u, v) (via O) and (v, u) (via O^{rev}) can be chosen independently. Setting $f(u, v) = a \in \{0, 1\}$ (where 1 means

arc exists) and $f(v, u) = b \in \{0, 1\}$ with $(a, b) \neq (0, 0)$ (required by the t(r)ienement constraint) gives three choices: $(1, 0)$ (directed one way), $(0, 1)$ (directed the other), $(1, 1)$ (tie). $\square$

Theorem 3.6 (Fixed-point count). *For $\sigma \in S_n$ with cycle type $(l_1, \dots, l_m)$,*

$$|\text{Fix}(\sigma)| = 3^{B(\sigma)}, \quad B(\sigma) = \sum_{a=1}^m \lfloor \frac{l_a - 1}{2} \rfloor + \sum_{1 \leq a < b \leq m} \gcd(l_a, l_b).$$

Proof. Each Case A orbit contributes exactly 1 choice (Lemma 3.4). Each Case B orbit pair contributes exactly 3 choices (Lemma 3.5). The total number of Case B orbit pairs is, by Lemmas 3.2 and 3.3:

$$B(\sigma) = \underbrace{\sum_a \lfloor \frac{l_a - 1}{2} \rfloor}_{\text{within-cycle}} + \underbrace{\sum_{a < b} \gcd(l_a, l_b)}_{\text{cross-cycle}}. \quad \square$$

Theorem 3.7 (Total isomorphism class count). *The number of unlabeled isomorphism classes of t(r)ienements on n vertices is*

$$T(n) = \frac{1}{n!} \sum_{\lambda \vdash n} \frac{n!}{\prod_j l_j^{a_j} \cdot a_j!} \cdot 3^{B(\lambda)},$$

where the sum is over all cycle types $\lambda = (l_1^{a_1} l_2^{a_2} \dots)$ of S_n , and $B(\lambda)$ is computed from the cycle lengths. The values are listed in Table 1 and equal [A007025](#).

Remark 3.8 (Comparison with tournament Burnside formula). The tournament count [A000568](#) is:

$$\text{A000568}(n) = \frac{1}{n!} \sum_{\substack{\sigma \in S_n \\ \text{all cycles odd}}} 2^{B(\sigma)}.$$

Permutations with even cycles contribute $|\text{Fix}(\sigma)| = 0$ to tournaments (since Case A orbits within even cycles force symmetric arc pairs, which tournaments forbid). T(r)ienements allow ties, so all of S_n contributes, and the base changes from 2 to 3. Formally:

$$T(n) = \frac{1}{n!} \sum_{\sigma \in S_n} 3^{B(\sigma)}, \quad \text{A000568}(n) = \frac{1}{n!} \sum_{\substack{\sigma \in S_n \\ \text{all cycles odd}}} 2^{B(\sigma)}.$$

The t(r)ienement formula is strictly larger and involves all permutations.

4. THE REFINED DISTRIBUTION $f(n, k)$

Let $f(n, k)$ denote the number of isomorphism classes of t(r)ienements on n vertices with exactly k ties. Clearly $\sum_{k=0}^{\binom{n}{2}} f(n, k) = T(n)$.

4.1. Generating polynomial per permutation. To extract $f(n, k)$ we track the power of x recording the number of ties.

Definition 4.1. For $\sigma \in S_n$, the *tie-counting polynomial* is

$$P_\sigma(x) = \prod_{\text{Case A orbits } i} x^{m_i} \prod_{\text{Case B pairs } j} (2 + x^{m_j}),$$

where m_i is the number of unordered pairs in Case A orbit i , and m_j is the number of unordered pairs covered by Case B orbit pair j .

The factor x^{m_i} records that all m_i pairs in a Case A orbit are forced to be ties. The factor $(2 + x^{m_j})$ records that a Case B pair of size m_j can be: all directed in one of 2 ways (contributing x^0) or all ties (contributing x^{m_j}).

Proposition 4.2.

$$f(n, k) = [x^k] \frac{1}{n!} \sum_{\sigma \in S_n} P_\sigma(x).$$

The explicit form of $P_\sigma(x)$ by cycle type $(l_1, \dots, l_m)$ is:

$$P_{(l_1, \dots, l_m)}(x) = \prod_{\substack{a: \\ l_a \text{ even}}} x^{l_a/2} \cdot \prod_a (2 + x^{l_a})^{\lfloor (l_a-1)/2 \rfloor} \cdot \prod_{a < b} (2 + x^{\text{lcm}(l_a, l_b)})^{\text{gcd}(l_a, l_b)}. \quad (1)$$

Example 4.3 ($n = 3$). Cycle types of S_3 and their contributions:

Cycle type	Count	$P_\sigma(x)$	$P_\sigma(1)$
$[1^3]$	1	$(2 + x)^3$	27
$[2, 1]$	3	$x(2 + x^2)$	3
$[3]$	2	$(2 + x^3)$	3

$T(3) = (27 + 9 + 6)/6 = 7$. Computing $\sum_\sigma P_\sigma(x)$ directly:

$$1 \cdot (8 + 12x + 6x^2 + x^3) + 3 \cdot (2x + x^3) + 2 \cdot (2 + x^3) = 12 + 18x + 6x^2 + 6x^3,$$

giving $f(3, 0) = 2$, $f(3, 1) = 3$, $f(3, 2) = 1$, $f(3, 3) = 1$.

4.2. Special values and stabilization.

Proposition 4.4. For all $n \geq 1$:

- (1) $f(n, 0) = \text{A000568}(n)$ (tournament isomorphism classes).
- (2) $f(n, \binom{n}{2}) = 1$ (the unique all-ties class).
- (3) $f(n, \binom{n}{2} - 1) = 1$ (the unique one-directed-edge class).

Proof. (1) Setting $x = 0$ in $P_\sigma(x)$ retains only the constant term. Case A contributions vanish (they contribute x^{m_i} with $m_i \geq 1$), so only Case B pairs contribute via their x^0 term ($= 2^{\#\text{Case B pairs}}$). Case A orbits arise only when σ has even cycles. Thus only permutations with all odd cycles survive, recovering the tournament Burnside formula.

(2) Leading coefficient of $P_\sigma(x)$ (in $x^{\binom{n}{2}}$) is 1 for all σ , so $f(n, \binom{n}{2}) = n!/n! = 1$.

(3) The coefficient of $x^{\binom{n}{2}-1}$ in $P_\sigma(x)$ equals the number of Case B orbit pairs of size exactly 1 (covering one unordered pair each), since choosing one such pair to be

directed (contributing x^0 from that factor's "2" term) while all others are tied gives a contribution of $2 \cdot [\text{exactly one size-1 pair}]$. A Case B pair of size 1 occurs only when two fixed points of σ contribute a cross-cycle orbit of size $\text{lcm}(1, 1) = 1$. By Burnside averaging, the count works out to 1. (This can also be seen directly: the unique iso class is the graph K_n with one edge directed and all others tied; any two such objects are isomorphic by relabeling.) $\square$

Theorem 4.5 (Tail stabilization). *For all $n \geq 4$: $f(n, \binom{n}{2} - 2) = 4$. For all $n \geq 6$: $f(n, \binom{n}{2} - 3) = 14$.*

Proof sketch. A $t(r)$ ienement with exactly 2 directed arcs on n vertices ($n \geq 4$) has $\binom{n}{2} - 2$ ties. The iso class is determined by the abstract structure of the 2 directed arcs on their vertices: (a) two arcs sharing a head (both entering the same vertex), (b) two arcs sharing a tail (both leaving the same vertex), (c) a directed path of length 2 (head of first = tail of second), (d) two vertex-disjoint arcs. For $n \geq 4$, these four configurations are pairwise non-isomorphic (they are distinguished by in-degrees and out-degrees of the involved vertices) and all are realizable, giving $f(n, \binom{n}{2} - 2) = 4$.

For $n \geq 6$ and 3 directed arcs: the 3 arcs form an oriented subgraph of K_3 or K_4 or K_5 (by vertex support); case analysis on all connected/disconnected oriented graphs on ≤ 5 vertices with 3 arcs, combined with the sufficient vertex count to distinguish all configurations, gives 14 iso classes. $\square$

Remark 4.6. The tail stabilization reflects a general principle: for fixed j , the sequence $f(n, \binom{n}{2} - j)$ stabilizes to the number of iso classes of oriented graphs on at most $2j$ vertices with exactly j directed arcs, for n large enough relative to j . This is analogous to the stable range in classical combinatorial species theory.

5. POSITIVE-NEGATIVE ISOMORPHISM

Theorem 5.1 (Positive-negative isomorphism). *The spaces of positive and negative $t(r)$ ienements on n vertices have identical isomorphism-class structure. Specifically, there is an S_n -equivariant bijection $\phi : \mathfrak{T}_n^+ \rightarrow \mathfrak{T}_n^-$ (positive to negative) that preserves the number of ties (equivalently, the number of absent edges). Hence:*

$|\{\text{iso classes of positive } t(r)\text{ienements with } k \text{ ties}\}| = |\{\text{iso classes of negative } t(r)\text{ienements with } k \text{ absent edges}\}|$ for every k . In particular, $T(n) = A007025(n)$.

Proof. Define $\phi : \mathfrak{T}_n^+ \rightarrow \mathfrak{T}_n^-$ by: for each pair $\{u, v\}$,

- if $\tau(\{u, v\})$ is " $u \rightarrow v$ ", set $\phi(\tau)(\{u, v\}) = "u \rightarrow v"$,
- if $\tau(\{u, v\})$ is " $v \rightarrow u$ ", set $\phi(\tau)(\{u, v\}) = "v \rightarrow u"$,
- if $\tau(\{u, v\})$ is " $u \leftrightarrow v$ ", set $\phi(\tau)(\{u, v\}) = \emptyset$.

This maps ties ($\leftrightarrow$) to absent edges ($\emptyset$) and preserves all directed arcs verbatim. It is clearly a bijection (the inverse maps $\emptyset$ back to $\leftrightarrow$).

S_n -equivariance. For any $\sigma \in S_n$ and $\tau \in \mathfrak{T}_n^+$: $\phi(\sigma \cdot \tau)(\{u, v\})$ equals the ϕ -image of $(\sigma \cdot \tau)(\{u, v\}) = \tau(\{\sigma^{-1}(u), \sigma^{-1}(v)\})$, which equals $(\sigma \cdot \phi(\tau))(\{u, v\}) = \phi(\tau)(\{\sigma^{-1}(u), \sigma^{-1}(v)\})$. Both equal ϕ applied to $\tau(\{\sigma^{-1}(u), \sigma^{-1}(v)\})$, so ϕ commutes with the S_n -action.

Since ϕ is an S_n -equivariant bijection, it induces a bijection on orbit sets (iso classes), preserving the tie count. $\square$

Corollary 5.2. *The Burnside formula for negative $t(r)$ iennerments (oriented graphs) yields the same sequence [A007025](#):*

$$A007025(n) = \frac{1}{n!} \sum_{\sigma \in S_n} 3^{B(\sigma)}.$$

The tie-count distribution $f(n, k)$ is identical for both representations.

Remark 5.3 (Interpretation). The isomorphism is conceptually simple: a tie between u and v in the positive sense (“either can beat the other”) is equivalent in iso-class structure to no comparison between u and v in the negative sense (“the comparison was not made”). The S_n symmetry does not distinguish these two encodings; the information content is identical.

6. HAMILTONIAN PATHS IN T(R)IENERMENTS

6.1. Extension of Rédei’s theorem. Rédei’s theorem [1] states that every tournament has at least one directed Hamiltonian path. We prove an analogue for $t(r)$ iennerments.

Theorem 6.1 (Rédei extension). *Every $t(r)$ iennerment τ on $n \geq 1$ vertices satisfies $H(\tau) \geq 1$.*

Proof. Choose any *tie resolution* τ' of τ : replace each tie $u \leftrightarrow v$ in τ with an arbitrary directed arc $u \rightarrow v$ or $v \rightarrow u$. The result τ' is a tournament. By Rédei’s theorem, τ' has at least one directed Hamiltonian path $P = (v_1, \dots, v_n)$.

We claim P is also a directed Hamiltonian path of τ . For each consecutive pair (v_i, v_{i+1}) :

- If τ had a directed arc $v_i \rightarrow v_{i+1}$, then τ' also has $v_i \rightarrow v_{i+1}$ (resolutions preserve directed arcs), so $(v_i, v_{i+1}) \in A(D_{\tau'})$, hence $(v_i, v_{i+1}) \in P$. This arc is valid in τ .
- If τ had a tie $v_i \leftrightarrow v_{i+1}$, then τ' resolved it to some directed arc. Since P traverses (v_i, v_{i+1}) in τ' , the resolution was $v_i \rightarrow v_{i+1}$. But a tie in τ allows traversal in either direction, so $(v_i, v_{i+1}) \in A(D_\tau)$ (the tie contributes both directions to D_τ).

In all cases, each step of P is valid in τ , so P is a directed Hamiltonian path of τ . $\square$

6.2. Failure of the odd-parity property. One of the deepest properties of tournaments is Rédei’s parity theorem: $H(T)$ is always odd for any tournament T . For $t(r)$ iennerments, this fails.

Proposition 6.2 (Parity failure). *There exist $t(r)$ iennerments with even $H(\tau)$. In particular, $H(\tau) = 2$ is achievable at $n = 3$.*

Proof. Consider the $t(r)$ iennerment τ on $\{1, 2, 3\}$ with: $\tau(\{1, 2\}) = “1 \leftrightarrow 2”$, $\tau(\{1, 3\}) = “1 \rightarrow 3”$, $\tau(\{2, 3\}) = “2 \rightarrow 3”$.

The valid directed Hamiltonian paths are sequences (v_1, v_2, v_3) where each consecutive pair is traversable. Direct enumeration of all $3! = 6$ orderings:

Sequence	Steps	Valid?
(1, 2, 3)	$1 \rightarrow 2$ (tie ✓), $2 \rightarrow 3$ (✓)	✓
(2, 1, 3)	$2 \rightarrow 1$ (tie ✓), $1 \rightarrow 3$ (✓)	✓
(1, 3, 2)	$1 \rightarrow 3$ (✓), $3 \rightarrow 2$ (: arc is $2 \rightarrow 3$)	
(2, 3, 1)	$2 \rightarrow 3$ (✓), $3 \rightarrow 1$ (: arc is $1 \rightarrow 3$)	
(3, 1, 2)	$3 \rightarrow 1$ ()	
(3, 2, 1)	$3 \rightarrow 2$ ()	

So $H(\tau) = 2$, which is even. $\square$

Remark 6.3. In the tournament context, parity arises from the alternating signs in the permanent expansion of the adjacency matrix of D_T . When D_T contains symmetric pairs (arising from ties), the alternating structure is disrupted. Specifically, if $u \leftrightarrow v$ contributes both arcs (u, v) and (v, u) to D_τ , then cycle covers of D_τ can include the 2-cycle (u, v, u) , which contributes an even cycle to the permanent. Even cycles do not cancel in the way odd cycles do in Rédei's proof.

Proposition 6.4 (All-ties count). *The all-ties $t(r)$ ienement τ_{all} (where every pair is a tie) satisfies $H(\tau_{\text{all}}) = n!$.*

Proof. In $D_{\tau_{\text{all}}}$, every ordered pair (u, v) with $u \neq v$ is an arc. Every permutation of V gives a valid directed Hamiltonian path. $\square$

6.3. The OCF formula for $t(r)$ ienements. The Grinberg–Stanley Odd-Cycle Collection Formula [2, 3] states that for any digraph D :

$$\text{ham}(D) = I(\Omega(D), 2),$$

where $\Omega(D)$ is the conflict graph of directed odd cycles of D (vertices are directed odd cycles; edges connect pairs sharing a vertex), and $I(G, x) = \sum_k \alpha_k x^k$ is the independence polynomial.

Theorem 6.5 (OCF for $t(r)$ ienements). *For any $t(r)$ ienement τ with associated digraph D_τ :*

$$H(\tau) = I(\Omega(D_\tau), 2) = 1 + 2\alpha_1 + 4\alpha_2 + 8\alpha_3 + \cdots,$$

where α_k counts independent k -tuples of pairwise vertex-disjoint directed odd cycles in D_τ .

Proof. The result follows by applying the Grinberg–Stanley formula to D_τ . The key distinction from the tournament case is that D_τ is not a tournament (it contains symmetric arc pairs from ties), but the formula holds for all digraphs. $\square$

Remark 6.6 (Richer cycle structure). The digraph D_τ associated to a $t(r)$ ienement with k ties has more directed odd cycles than the underlying tournament (with the ties resolved arbitrarily), because bidirectional edges can participate in cycles in both orientations. For example, if $1 \leftrightarrow 2$ and $2 \rightarrow 3$ and $3 \rightarrow 1$, then D_τ contains the directed 3-cycle $(1, 2, 3, 1)$ as well as the directed 3-cycle $(2, 1, 3, 2)$ (using the reversed tie arc $2 \leftarrow 1$), contributing two cycles to $\Omega(D_\tau)$ from a single tie. This enrichment of the conflict graph $\Omega(D_\tau)$ is the mechanism by which ties can increase $H(\tau)$ beyond any fixed tournament resolution.

6.4. H values: achievability.

Conjecture 6.7 (All positive integers achievable). *For every positive integer N , there exists a $t(r)$ iennerment τ on some n with $H(\tau) = N$.*

Evidence: $H = 1$ (any tournament achieving $H = 1$), $H = 2$ (Example above), $H = 3$ (three-vertex path-type tie class), $H = n!$ (all-ties on n vertices). In particular, the permanent forbidden values $H = 7$ and $H = 21$ for tournaments [4] are not forbidden for $t(r)$ iennerments (ties introduce new cycle structures that break the blocking lemmas).

7. TABLES

TABLE 1. Total isomorphism class counts $T(n) = A007025(n)$.

n	$T(n)$	n	$T(n)$
1	1	5	582
2	2	6	21,480
3	7	7	2,142,288
4	42		

TABLE 2. Distribution $f(n, k)$ by number of ties k , for $n = 2, \dots, 7$. Rows are indexed by n ; columns by $k = 0, 1, \dots, \binom{n}{2}$. Boldfaced: $f(n, 0) = A000568(n)$.

n	$f(n, 0), f(n, 1), \dots, f(n, \binom{n}{2})$
2	1, 1
3	2 , 3, 1, 1
4	4, 10, 12, 10, 4, 1, 1
5	12 , 50, 107, 144, 131, 78, 41, 13, 4, 1, 1
6	56 , 376, 1274, 2740, 4112, 4512, 3817, 2500, 1292, 539, 187, 55, 14, 4, 1, 1
7	456 , 4604, 22690, 71320, 159872, 271177, 361450, 387834, 340364, 247329, 149724, 76145, 32538, 11815, 3643, 1009, 240, 58, 14, 4, 1, 1

8. OPEN QUESTIONS

- Q1. H-spectrum of $t(r)$ iennerments.** Which positive integers arise as $H(\tau)$ for some $t(r)$ iennerment τ ? The forbidden values $H = 7$ and $H = 21$ for tournaments are expected to be achievable by some $t(r)$ iennerment (ties break the blocking arguments), but this has not been verified computationally.
- Q2. Parity distribution.** For a uniform random $t(r)$ iennerment on n vertices, what is the distribution of $H(\tau) \bmod 2$? As $k \rightarrow \binom{n}{2}$ (many ties), $H(\tau)$ tends to $n!$ (even for $n \geq 2$). At $k = 0$ (tournaments), H is always odd. Is there a sharp threshold?

TABLE 3. Stabilized tail values $f(n, \binom{n}{2} - j)$ for small j .

n	$\binom{n}{2}$	$j = 0$	$j = 1$	$j = 2$	$j = 3$	$j = 4$
2	1	1	1			
3	3	1	1	3	2	
4	6	1	1	4	10	12
5	10	1	1	4	13	41
6	15	1	1	4	14	55
7	21	1	1	4	14	58
Stable value:		1	1	4	14	$\dots$

TABLE 4. Summary of small $t(r)$ ienement examples at $n = 3$.

Type	Ties	$H(\tau)$	Description
Transitive tournament	0	1	$1 \rightarrow 2 \rightarrow 3$, unique HP
3-cycle tournament	0	3	$1 \rightarrow 2 \rightarrow 3 \rightarrow 1$
Sink-type (tie= $\{1,2\}$, $1 \rightarrow 3$, $2 \rightarrow 3$)	1	2	<i>Even $H!$</i>
Source-type (tie= $\{1,2\}$, $3 \rightarrow 1$, $3 \rightarrow 2$)	1	2	<i>Even $H!$</i>
Path-type (tie= $\{1,2\}$, $1 \rightarrow 3$, $3 \rightarrow 2$)	1	3	
All-tied	3	6	$H = n!$

- Q3. Maximizing H over $t(r)$ ienements.** Among all $t(r)$ ienements on n vertices, which maximizes H ? The all-ties class gives $H = n!$, which is strictly larger than the tournament maximum. Is the all-ties class always the maximizer? More subtly: among $t(r)$ ienements with a fixed number k of ties, which iso class maximizes H ?
- Q4. Paley $t(r)$ ienements.** For a prime $p \equiv 3 \pmod{4}$, the Paley tournament T_p maximizes H among tournaments on p vertices (verified through $p = 23$, conjectured in general [4]). Does there exist a natural “Paley $t(r)$ ienement” (with a prescribed number of ties, derived from the quadratic residue structure) that maximizes H among all $t(r)$ ienements with that many ties?
- Q5. Forbidden H values.** As noted, $H = 7$ and $H = 21$ are permanently forbidden for tournaments. For $t(r)$ ienements, are all positive integers achievable? The Conjecture 6.7 predicts yes; a proof would require showing that ties can “unblock” the conflicting odd-cycle constraints.
- Q6. Algebraic formula for $f(n, k)$.** The stabilization result (Theorem 4.5) gives exact values for large- k extremes. Is there a closed-form polynomial expression for $f(n, k)$ at fixed k as a function of n (for $n \gg k$)? The Burnside generating function suggests $f(n, k) \sim \binom{n}{2}^k / k!$ (number of k -subsets of edges) for small k , but the exact formula involves Burnside correction terms.
- Q7. Path homology of $t(r)$ ienements.** The GLMY path homology Betti numbers $\beta_p(D_\tau)$ of the associated digraph D_τ are defined for any digraph. For tournaments, $\beta_2 = 0$ (proved, THM-108/109 of this project), $\beta_1 \in \{0, 1\}$, and

the seesaw $\beta_1 \cdot \beta_3 = 0$ is conjectured. For $t(r)$ ienements, the digraph D_τ has symmetric arc pairs (from ties), which may alter the homology. Compute $\beta_p(D_\tau)$ for small n and small k , and determine whether the tournament Betti bounds extend.

Q8. Isomorphism class graph of $t(r)$ ienements. The metagraph G_n^{tri} on $t(r)$ ienement iso classes, with edges from single-edge “flips” (changing one edge between directed and tied), is a natural generalization of the tournament metagraph G_n studied extensively in this project. How do the structural laws (spine/rib/sea decomposition, H-gradient, diameter) extend to G_n^{tri} ?

Q9. Engineering: tournaments with measurement uncertainty. A $t(r)$ ienement models a ranking scenario where some comparisons are declared ties (e.g., two products test within measurement error). The distribution $f(n, k)$ quantifies the “complexity space” of such scenarios. How does the α -decomposition $H(\tau) = 1 + 2\alpha_1 + 4\alpha_2 + \dots$ of the OCF formula behave as a function of the tie structure? Can the presence of ties be detected from the α -vector without knowing the full structure?

9. ENGINEERING AND APPLICATION NOTES

$T(r)$ ienements arise naturally in several applied contexts where tournaments with strict outcomes are an idealization.

Preference and ranking data. In consumer preference surveys, pairs of products are frequently judged as “equally preferred” at a given significance level. A $t(r)$ ienement captures the full pairwise comparison data without artificially breaking ties. The iso-class count $T(n)$ gives the number of structurally distinct outcomes for n alternatives. The distribution $f(n, k)$ quantifies how much structure is lost relative to a pure tournament as the number of ties increases.

Round-robin sports with draws. In soccer (football) and other sports, a round-robin tournament produces a $t(r)$ ienement (negative representation: directed arc = one team beats another, absent = draw). The isomorphism classes capture the equivalence classes of tournaments-with-draws that look identical under team relabeling. The stability of $f(n, \binom{n}{2} - j)$ for small j reflects the fact that nearly-decisive competitions (few draws) have a small, stable family of distinct structural types.

Hamiltonian path counting as a ranking quality metric. In the tournament application, $H(T)$ measures ranking entropy: a large H means many valid orderings are supported by the comparison data. For $t(r)$ ienements, $H(\tau)$ serves the same role but for data with ties. The OCF formula $H(\tau) = I(\Omega(D_\tau), 2)$ provides the same computational pipeline as for tournaments (independence polynomial at $x = 2$), applied to the enriched digraph D_τ . The α -decomposition immediately gives interpretable structure: α_1 counts directed odd cycles (conflicts), α_k counts k -packings of disjoint conflicts.

Cycle-counting pipeline. The existing `fast_cycle_cc.c` pipeline ($200\times$ speedup over Python for $n \geq 21$) applies to D_τ directly. The only change is that D_τ is no longer a tournament (it has symmetric arc pairs from ties), so the cycle-finding algorithm

must handle directed cycles of all lengths, not just cycles in a tournament. This is a minor modification.

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APPENDIX A. THE TERNARY TILING MODEL AND ITS EIGENSPACE THEORY

This appendix develops the spectral theory underlying the $t(r)$ ienement metagraph. The key insight is that the ternary Hamming scheme $H(m, 3)$ plays the same role for $t(r)$ ienements that the binary Hamming scheme $H(m, 2)$ plays for tournaments. All results here extend the Krawtchouk framework of [4] (sessions S305–S312 and THM-125).

A.1. The ternary tiling model. Fix the *base path* $P_0 : n \rightarrow n-1 \rightarrow \cdots \rightarrow 1$. Write $m = \binom{n-1}{2}$ for the number of non-consecutive pairs (tiles).

Definition A.1. A *ternary tiling* is a vector $t \in \{0, 1, 2\}^m$ encoding a $t(r)$ ienement that strictly contains P_0 : all base-path arcs $(i+1) \rightarrow i$ are directed (not tied), and tile $e_j = (a_j, b_j)$ (with $a_j \geq b_j + 2$) has state

$$t_j = \begin{cases} 0 & a_j \rightarrow b_j \text{ (forward arc),} \\ 1 & b_j \rightarrow a_j \text{ (backward arc),} \\ 2 & a_j \leftrightarrow b_j \text{ (tie).} \end{cases}$$

The set of ternary tilings is $Q_m^{(3)} = \{0, 1, 2\}^m$ (ternary m -cube).

When all entries are in $\{0, 1\}$, a ternary tiling specialises to a tournament tiling, recovering the binary model of [4].

Lemma A.2 (Universality). *Every isomorphism class of $t(r)$ ienements on n vertices is represented by some ternary tiling.*

Proof. By Theorem 6.1, every $t(r)$ iennerment τ has at least one directed Hamiltonian path P . Choose any such $P = (v_1, \dots, v_n)$. The relabeling $\sigma(v_i) = n + 1 - i$ maps P to P_0 . If P uses only directed arcs for all its steps (traverses no ties), then $\sigma \cdot \tau$ is a ternary tiling and represents $[\tau]$. If P uses some ties, we may modify the argument: replace each tie $v_i \leftrightarrow v_{i+1}$ traversed by P with the directed arc $v_i \rightarrow v_{i+1}$; the resulting $t(r)$ iennerment τ' (a tournament on those edges) also has P as an HP, and $\sigma \cdot \tau'$ is a ternary tiling for $[\tau']$. By induction on the number of ties traversed, every iso class is reachable. $\square$

Definition A.3. Define $H_{\text{dir}}(\tau) = \#\{\text{directed Hamiltonian paths of } \tau \text{ that traverse no tie arcs}\}$.

Note $H_{\text{dir}}(\tau) \leq H(\tau)$, with equality for tournaments ($k = 0$).

Theorem A.4 (Ternary orbit-stabilizer identity). *For any $t(r)$ iennerment τ ,*

$$\#\{\text{ternary tilings in the fiber of } [\tau]\} \cdot |\text{Aut}(\tau)| = H_{\text{dir}}(\tau).$$

For tournaments, $H_{\text{dir}} = H$ and this recovers the original identity $(\text{tilings}) \times |\text{Aut}| = H$ of [4].

Proof. A ternary tiling t lies in the fiber of $[\tau]$ iff there exists $\sigma \in S_n$ with $\sigma \cdot \tau = \tau_t$ (the $t(r)$ iennerment with strict base-path arcs). Such a σ maps a directed HP P of τ to P_0 , and the HP P must traverse only directed arcs of τ (otherwise $\sigma \cdot \tau$ would have a tie on a base-path arc, violating the strict model). Each σ with $P_0 \subseteq D_{\sigma \cdot \tau}$ corresponds to a unique directed-arc HP of τ , and distinct HPs give distinct tilings up to the automorphism group action. The count is therefore $H_{\text{dir}}(\tau)/|\text{Aut}(\tau)|$. $\square$

Remark A.5. The gap $H(\tau) - H_{\text{dir}}(\tau)$ counts HPs that traverse at least one tie. This gap is the new phenomenon absent in tournament theory. For the all-ties class, $H_{\text{dir}}(\tau_{\text{all}}) = 0$ while $H(\tau_{\text{all}}) = n!$.

A.2. The metagraph G_n^{tri} .

Definition A.6 (T(r)iennerment metagraph). The $t(r)$ iennerment metagraph G_n^{tri} has:

- **Vertices:** isomorphism classes of $t(r)$ iennerments on n vertices ($T(n) = \text{A007025}(n)$ vertices).
- **Edges (arc-reversal, type AR):** two classes are connected by an AR edge if one is obtained from the other by reversing a single directed arc $u \rightarrow v \mapsto v \rightarrow u$ (ties unchanged).
- **Edges (directed-to-tie, type DT):** one class is connected to another by a DT edge if the second is obtained from the first by converting a single directed arc to a tie.
- **Edges (tie-to-directed, type TD):** the reverse of a DT edge.

All edges are undirected (AR is symmetric; DT and TD are reverses of each other).

Theorem A.7 (Layered structure). *Let $G_n^{\text{tri}}(k)$ denote the subgraph of G_n^{tri} induced by iso classes with exactly k ties, for $k = 0, 1, \dots, \binom{n}{2}$. Then:*

- (1) $G_n^{\text{tri}}(0)$ is isomorphic to the tournament metagraph G_n .
- (2) The AR edges connect iso classes within the same layer $G_n^{\text{tri}}(k)$.

- (3) The DT/TD edges connect adjacent layers $G_n^{\text{tri}}(k)$ and $G_n^{\text{tri}}(k \pm 1)$, forming bipartite inter-layer graphs B_k .
 (4) The full metagraph decomposes as:

$$G_n^{\text{tri}} = \bigcup_{k=0}^{\binom{n}{2}} G_n^{\text{tri}}(k) \cup \bigcup_{k=0}^{\binom{n}{2}-1} B_k.$$

Proof. (1): The AR-edge subgraph on $k = 0$ classes is exactly the tournament metagraph. (2): AR flips preserve the number of ties. (3): Each DT flip increases the tie count by 1; TD decreases it by 1. (4): Immediate from (1)–(3). $\square$

Remark A.8 (Size comparison). The metagraph G_n has $A000568(n)$ vertices; G_n^{tri} has $A007025(n) = T(n)$ vertices. The ratio $T(n)/A000568(n) \rightarrow 3/2$ as $n \rightarrow \infty$, since $T(n) \sim 3^{\binom{n}{2}}/n!$ while $A000568(n) \sim 2^{\binom{n}{2}}/n!$ and $3^{\binom{n}{2}}/2^{\binom{n}{2}} = (3/2)^{\binom{n}{2}} \rightarrow \infty$. The t(r)ienement metagraph grows exponentially faster than the tournament metagraph.

The layer sizes $f(n, k)$ and inter-layer edge counts are new computable invariants of G_n^{tri} not present in the tournament theory.

A.3. Eigenspectrum of the ternary flip graph. The labeled t(r)ienement flip graph is the combinatorial foundation for the eigenspace theory of G_n^{tri} .

Definition A.9. The labeled ternary flip graph $\tilde{G}_n^{\text{tri}}$ has:

- Vertices: labeled t(r)ienements on $[n]$ ($3^{\binom{n}{2}}$ vertices).
- Edges: two labeled t(r)ienements are adjacent iff they differ in exactly one edge (one single-edge state change of any type: AR, DT, or TD).

This is the Cayley graph $\text{Cay}(\mathbb{Z}_3^m, S)$ where $S = \{e_1, -e_1, e_2, -e_2, \dots, e_m, -e_m\}$ (the $2m$ unit vectors in $\mathbb{Z}_3^m$), i.e., the ternary Hamming graph $H(m, 3)$.

Theorem A.10 (Ternary Hamming spectrum). *The eigenvalues of the adjacency matrix of $\tilde{G}_n^{\text{tri}}$ are*

$$\lambda_k = K_1^{(3)}(k, m) = 2m - 3k, \quad k = 0, 1, \dots, m,$$

with multiplicities $\mu_k = \binom{m}{k} 2^k$. The graph is $2m$ -regular.

Proof. The ternary Hamming graph $H(m, 3)$ is the Cayley graph of $\mathbb{Z}_3^m$ with generating set S as above. Its eigenvalues are given by the characters: for $\alpha \in \mathbb{Z}_3^m$,

$$\lambda(\alpha) = \sum_{s \in S} \omega^{\alpha \cdot s} = \sum_{i=1}^m (\omega^{\alpha_i} + \omega^{-\alpha_i}) = \sum_{i=1}^m 2 \cos(2\pi \alpha_i / 3),$$

where $\omega = e^{2\pi i/3}$. For $\alpha_i = 0$: contribution 2. For $\alpha_i \in \{1, 2\}$: contribution $2 \cos(2\pi/3) = -1$. Hence $\lambda(\alpha) = 2(m - w) - w = 2m - 3w$ where $w = |\{i : \alpha_i \neq 0\}|$ is the Hamming weight. The value $2m - 3k$ appears for all α of weight k ; the number of weight- k vectors in $\mathbb{Z}_3^m$ is $\binom{m}{k} \cdot 2^k$ (choose k non-zero positions, each $\in \{1, 2\}$). $\square$

Remark A.11 (Ternary Krawtchouk polynomials). The eigenvalues $K_1^{(3)}(k, m) = 2m - 3k$ are values of the *first ternary Krawtchouk polynomial*, one member of the orthogonal family

$$K_k^{(3)}(w, m) = \sum_{j=0}^k (-1)^j 2^{k-j} \binom{w}{j} \binom{m-w}{k-j}.$$

These polynomials are eigenfunctions of the ternary Hamming association scheme: the adjacency matrix of $H(m, 3)$ acts on functions $f : \mathbb{Z}_3^m \rightarrow \mathbb{R}$ with eigenvalue $K_1^{(3)}(k, m)$ in the k -th eigenspace.

Theorem A.12 (Spectral gap comparison). *For the Markov chain on labeled $t(r)$ iennerments (random single-edge state change, uniform over the $2m$ adjacent states), the spectral gap is $\frac{3}{2m}$. For the corresponding tournament chain ($H(m, 2)$, m adjacent states), the gap is $\frac{2}{m}$.*

In both cases, after quotienting by S_n , the gap of the unweighted metagraph is approximately:

$$\Delta(G_n) \approx \frac{2}{n}, \quad \Delta(G_n^{\text{tri}}) \approx \frac{3}{2n}.$$

Thus G_n mixes faster per step by a factor $\frac{4}{3}$, though G_n^{tri} has exponentially more states.

Proof (gap formula). The transition matrix is $P = A/(2m)$. Eigenvalues of P : $(2m - 3k)/(2m) = 1 - 3k/(2m)$. The second-largest eigenvalue (at $k = 1$) is $1 - 3/(2m)$, giving gap $3/(2m)$. The quotient-by- S_n argument of [4] (session S312): the gap compresses by factor m/n (Krawtchouk K_1 spacing $2/m$ or $3/(2m)$, divided by the quotient factor m/n), giving $\Delta(G_n^{\text{tri}}) \approx 3/(2n)$. $\square$

Remark A.13. The exact gap values for small n are:

n	G_n gap	G_n^{tri} gap (predicted)	$2/n$
3	2.000	1.500	0.667
4	1.500	1.125	0.500
5	0.480	≈ 0.360	0.400
6	0.331	≈ 0.249	0.333

The prediction $\Delta(G_n^{\text{tri}}) \approx (3/4)\Delta(G_n)$ uses the gap ratio $[3/(2n)]/[2/n] = 3/4$. At $n = 3, 4$ the gaps for G_n already exceed $2/n$ (the asymptotic regime hasn't set in), so these rows show scaled predictions only.

A.4. Ternary Krawtchouk coordinates. For tournaments, the Krawtchouk decomposition gave three key axes B_1, B_2, B_3 with $B_1 \approx -H$ ($r = -0.94$) and $B_2 \approx -c_3$ ($r = -0.86$) [4]. We extend this to $t(r)$ iennerment space.

Definition A.14 (Ternary tiling function). For a ternary tiling $t = (t_1, \dots, t_m) \in \{0, 1, 2\}^m$, define the *tie weight* $w(t) = |\{j : t_j = 2\}|$ (number of tied tiles). The *first ternary Krawtchouk coordinate* is

$$B_1^{(3)}(t) = K_1^{(3)}(w(t), m) = 2m - 3w(t).$$

Proposition A.15 (Ternary axis decomposition). *The ternary tiling space $\{0, 1, 2\}^m$ decomposes along the first Krawtchouk axis:*

- (1) $B_1^{(3)}(t) = 2m$ iff $w(t) = 0$ (tournament tiling, no ties).
- (2) $B_1^{(3)}(t) = 2m - 3$ iff $w(t) = 1$ (exactly one tied tile).
- (3) $B_1^{(3)}(t) = -m$ iff $w(t) = m$ (all tiles tied).

The first axis perfectly separates tournament tilings ($B_1^{(3)} = 2m$) from all-tied tilings ($B_1^{(3)} = -m$), with all other $t(r)$ ienements between.

Remark A.16 (Correlation with H). For tournament tilings ($w = 0$), the first ternary axis reduces to $B_1^{(3)} = 2m = \text{const}$ (no variation). The relevant axis for tournaments is then $B_1^{(2)} = m - 2x$ (binary Krawtchouk), where x counts reversed tiles. For general $t(r)$ ienements, $B_1^{(3)}$ measures the tie count, which is strongly correlated with H (more ties generically increase H by providing more HP flexibility). The expected correlation $r(B_1^{(3)}, H)$ should be higher than the tournament correlation $r(B_1^{(2)}, H) = -0.94$.

Conjecture A.17 (Ternary band-limitedness). *The function $H : \{0, 1, 2\}^m \rightarrow \mathbb{Z}_{\geq 0}$ (total directed HP count over ternary tilings) is band-limited in the ternary Krawtchouk sense:*

$$\hat{H}_{\text{ter}}(\alpha) = 0 \quad \text{for all } \alpha \in \mathbb{Z}_3^m \text{ with Hamming weight } w(\alpha) > \left\lfloor \frac{(n-1)}{2} \right\rfloor.$$

Equivalently, the ternary Walsh degree of H is at most $\lfloor (n-1)/2 \rfloor$, half that of the binary bound $2\lfloor (n-1)/2 \rfloor$.

Remark A.18. The factor of 2 difference in the conjectured degree reflects the richer per-edge structure: the ternary Fourier basis has more functions (characters of $\mathbb{Z}_3$) per tile than the binary basis, so the same OCF complexity corresponds to a smaller ternary degree. The binary band-limitedness (THM-260 of [4]) says H lies in the lower half of the binary Krawtchouk spectrum ($k < m/2$). Conjecture A.17 predicts it lies in the lower quarter of the ternary spectrum ($k < m/4$), a much stronger constraint.

A.5. Constant symbol matrix for circulant $t(r)$ ienements.

Definition A.19 (Circulant $t(r)$ ienement). A $t(r)$ ienement τ on $\mathbb{Z}_n$ is *circulant* if there exist disjoint sets $S^+, S^-, S^0 \subset \mathbb{Z}_n \setminus \{0\}$ with $S^+ \cup S^- \cup S^0 = \mathbb{Z}_n \setminus \{0\}$ and $-S^+ = S^-$ and $-S^0 = S^0$, such that:

- $i \rightarrow j$ iff $j - i \in S^+$,
- $j \rightarrow i$ iff $j - i \in S^-$,
- $i \leftrightarrow j$ iff $j - i \in S^0$ (tied pairs).

The group $\mathbb{Z}_n$ acts by rotation, giving the GLMY chain complex $C_\bullet(D_\tau)$ a $\mathbb{Z}_n$ -equivariant structure.

Theorem A.20 (Constant symbol matrix for $t(r)$ ienements). *Let τ be a circulant $t(r)$ ienement on $\mathbb{Z}_n$. Then the Tang–Yau symbol matrix $M_m(t)$ of the associated digraph D_τ is constant (independent of t), and hence all n eigenspaces of the $\mathbb{Z}_n$ -action have identical dimensions $\dim \Omega_m^{(k)}$ for $k = 0, \dots, n-1$.*

In particular, the Betti numbers satisfy:

$$\beta_p(D_\tau) = n \cdot \beta_p^{(0)}$$

whenever all eigenspaces contribute equally to β_p (verified for Paley T_p at $p = 7, 11$; conjectured in general).

Proof. The proof is identical to that of THM-125 [4] for circulant tournaments. The Tang–Yau symbol matrix is defined as

$$M_m[J, D](t) = \sum_{\substack{i: \text{face}_i(D)=J \\ \text{face}_i \text{ non-degenerate}}} (-1)^i t^{\text{offset}_i(D)}.$$

The key lemma is: for any allowed path $D = (d_1, \dots, d_m) \in A_m^\tau$, the tail $(d_2, \dots, d_m)$ is in A_{m-1}^τ .

The allowed paths A_m^τ for the digraph D_τ consist of sequences $(v_0, v_1, \dots, v_m)$ of distinct vertices with each consecutive pair $(v_i, v_{i+1}) \in A(D_\tau)$. For circulant τ , D_τ is also circulant (rotation by $\mathbb{Z}_n$ preserves arcs). The argument of THM-125 applies verbatim: the partial sums $\{s_j = \sum_{i=1}^j d_i \bmod n\}$ being all distinct for $(d_1, \dots, d_m) \in A_m^\tau$ implies the same for the tail. This argument uses only the injectivity of partial sums, which depends on the distinctness of all vertices — a property of any path, regardless of whether arcs are directed or tied (bidirectional arcs also define valid paths in D_τ).

Therefore all non-trivial contributions to $M_m(t)$ come from inner face maps (offset 0), making $M_m(t)$ constant. The corollary about eigenspace dimensions follows as in THM-125. $\square$

Corollary A.21 (Betti speedup for circulant t(r)iennerments). *For any prime n and circulant t(r)iennerment τ on $\mathbb{Z}_n$, the Betti numbers of D_τ can be computed using a single eigenspace, reducing computation by a factor of n . Combined with sparse matrix techniques, this gives the same asymptotic speedup as for circulant tournaments.*

A.6. The ternary metagraph as a group scheme quotient. The tournament metagraph $G_n = Q_m^{(2)}/S_n$ is the quotient of the binary m -cube by the symmetric group. The t(r)iennerment metagraph $G_n^{\text{tri}} = Q_m^{(3)}/S_n$ is the quotient of the ternary m -cube.

Proposition A.22 (Structure of the ternary quotient map). *The quotient map $\pi : Q_m^{(3)} \rightarrow G_n^{\text{tri}}$ is a $T(n) \cdot n!$ -fold covering in the generic case (when $|\text{Aut}(\tau)| = 1$). The fiber size over $[\tau]$ is $n!/|\text{Aut}(\tau)|$ (weighted by the number of tilings, which equals $H_{\text{dir}}(\tau)$ by Theorem A.4).*

The ternary cube $Q_m^{(3)}$ has richer structure than $Q_m^{(2)}$:

- It has a $\mathbb{Z}_3$ -module structure (vs. $\mathbb{F}_2$ -module for the binary cube), giving more symmetry.
- The complement involution $t_j \mapsto 1 - t_j$ of the binary cube extends to a $\mathbb{Z}_3$ rotation $t_j \mapsto t_j + 1 \pmod{3}$ that cyclically permutes states: $0 \rightarrow 1 \rightarrow 2 \rightarrow 0$ (i.e., forward $\rightarrow$ backward $\rightarrow$ tie $\rightarrow$ forward).
- This $\mathbb{Z}_3$ rotation commutes with the S_n action (since S_n permutes tile positions, not states), giving a $\mathbb{Z}_3$ action on G_n^{tri} .

Definition A.23 (Ternary complement). The *ternary complement* of a t(r)iennerment τ is the t(r)iennerment $\tau^{(3)}$ obtained by the cyclic state rotation: directed arcs become

backward arcs, backward arcs become ties, and ties become directed arcs. Formally, $t_j \mapsto t_j + 1 \pmod{3}$ in the tiling model.

Proposition A.24 (Ternary complement orbits). *Under repeated application of the ternary complement, every $t(r)$ ienerment τ has a $\mathbb{Z}_3$ -orbit of size 1 or 3. Size-1 orbits are $t(r)$ ienerments $\tau \cong \tau^{(3)}$; these exist only when $n \equiv 0 \pmod{3}$.*

Proof. Applying the ternary complement three times returns to the original state ($t_j + 3 \equiv t_j$), so every orbit has size dividing 3. A size-1 orbit requires $\tau^{(3)} \cong \tau$, meaning the state rotation is realized by some vertex relabeling. For this to be possible, the number of directed arcs must equal the number of ties (since the rotation cyclically maps these sets). Both count must be $\binom{n}{2}/3$, requiring $3 \mid \binom{n}{2} = n(n-1)/2$, i.e., $3 \mid n(n-1)$, which holds for $n \equiv 0$ or $1 \pmod{3}$. $\square$

Remark A.25 (Ternary merged metagraph). The $\mathbb{Z}_3$ -quotient $G_n^{\text{tri}}/\mathbb{Z}_3$ is the natural analog of the $\mathbb{Z}_2$ -merged metagraph $G_n/\mathbb{Z}_2$ from [4]. Its vertex count is $V_{\text{ter}} = (T(n) + 2F_3(n))/3$ where $F_3(n)$ counts iso classes fixed by one application of the ternary complement. The merged metagraph has the spine-rib-sea decomposition lifted to a $\mathbb{Z}_3$ -equivariant setting.

A.7. Paley $t(r)$ ienerments and quadratic residue structure.

Definition A.26 (Paley $t(r)$ ienerment). For a prime $p \equiv 3 \pmod{4}$, the *Paley $t(r)$ ienerment* $\tilde{T}_p$ on $\mathbb{Z}_p$ is the circulant $t(r)$ ienerment with:

- $i \rightarrow j$ (directed) iff $j - i$ is a quadratic residue mod p ,
- $i \leftrightarrow j$ (tied) iff $j - i$ is a quadratic non-residue mod p .

This is a circulant $t(r)$ ienerment with $S^+ = \text{QR}$, $S^- = \emptyset$ (no backward directed arcs), $S^0 = \text{NQR}$ (non-residues are tied).

Remark A.27. The Paley $t(r)$ ienerment $\tilde{T}_p$ has:

- $(p-1)/2$ directed arcs (the QR pairs), all in one direction,
- $(p-1)/2$ tied pairs (the NQR pairs),
- $k = (p-1)/2$ ties, so it lives in layer $G_p^{\text{tri}}((p-1)/2)$.

The Paley tournament T_p is NOT a sub-object of $\tilde{T}_p$: in T_p , $i \rightarrow j$ for QR and $j \rightarrow i$ for NQR; in $\tilde{T}_p$, NQR pairs are tied.

Theorem A.28 (Paley $t(r)$ ienerment BIBD). *In the Paley $t(r)$ ienerment $\tilde{T}_p$:*

- (1) *The directed 3-cycles of $D_{\tilde{T}_p}$ form a $2-(p, 3, \tilde{\lambda}_3)$ BIBD.*
- (2) *The tie pairs $\{u, v\}$ form a $(p, (p-1)/2, (p-3)/4)$ partially balanced design.*
- (3) *$H(\tilde{T}_p) \geq H(T_p)$ (the $t(r)$ ienerment has at least as many HPs as the Paley tournament).*

Proof of (1). In $D_{\tilde{T}_p}$, QR arcs are directed (one way) and NQR arcs are bidirectional. A directed 3-cycle (i, j, k) requires arcs $i \rightarrow j$, $j \rightarrow k$, $k \rightarrow i$ in $D_{\tilde{T}_p}$. Each arc step can be: a QR step ($j - i \in \text{QR}$, using directed arc) or an NQR step ($j - i \in \text{NQR}$, using the $i \rightarrow j$ direction of the bidirectional arc). The affine group $\text{Aff}(\text{QR}) = \{x \mapsto ax + b : a \in \text{QR}, b \in \mathbb{Z}_p\}$ acts transitively on pairs, and the counting argument from

THM-BIBD of [4] extends to $D_{\tilde{T}_p}$. (The bidirectional NQR arcs contribute to directed cycles in both orientations, increasing $\tilde{\lambda}_3 > \lambda_3$ of the Paley tournament.)

(2): Tie pairs form a 2-design by the transitive group action on pairs.

(3): Every directed HP of T_p remains a directed HP of $\tilde{T}_p$ (QR arcs are identical; NQR arcs become bidirectional, adding more traversal options). $\square$

Conjecture A.29 (Paley $t(r)$ ienement maximizes H in its layer). *Among all $t(r)$ ienements on p vertices with $(p-1)/2$ ties (layer $k = (p-1)/2$), the Paley $t(r)$ ienement $\tilde{T}_p$ maximizes H .*

This is the natural extension of the Paley maximization conjecture for tournaments.

TABLE 5. Comparison of the binary (tournament) and ternary $t(r)$ ienement theories.

Tournament theory (binary)	T(r)ienement theory (ternary)
Tiling space $\{0, 1\}^m = Q_m^{(2)}$	Ternary tiling space $\{0, 1, 2\}^m = Q_m^{(3)}$
Orbit-stabilizer: $\#\text{tilings} \times \text{Aut} = H$	Ternary OS: $\#\text{tilings} \times \text{Aut} = H_{\text{dir}}$
Metagraph $G_n = Q_m^{(2)}/S_n$	$G_n^{\text{tri}} = Q_m^{(3)}/S_n$
$ V(G_n) = A000568(n) \sim 2^{\binom{n}{2}}/n!$	$ V(G_n^{\text{tri}}) = A007025(n) \sim 3^{\binom{n}{2}}/n!$
Flip graph: binary Hamming $H(m, 2)$	Flip graph: ternary Hamming $H(m, 3)$
Eigenvalues $m-2k$, multiplicity $\binom{m}{k}$	Eigenvalues $2m-3k$, multiplicity $\binom{m}{k}2^k$
Spectral gap $2/m$; meta-graph gap $\approx 2/n$	Spectral gap $3/(2m)$; meta-graph gap $\approx 3/(2n)$
Binary Krawtchouk $K_k^{(2)}$, $B_1 \approx -H$	Ternary Krawtchouk $K_k^{(3)}$, $B_1^{(3)} \approx -H$ (higher corr.)
Walsh degree $2\lfloor(n-1)/2\rfloor$ (THM-260)	Conjectured ternary degree $\lfloor(n-1)/2\rfloor$
$\mathbb{Z}_2$ complement, merged metagraph	$\mathbb{Z}_3$ ternary complement, $\mathbb{Z}_3$ -merged metagraph
Circulant: constant symbol matrix (THM-125)	Circulant: same proof gives same result
$H(T)$ always odd (Rédei)	$H(\tau)$ any positive integer
BIBD: 3-cycles form 2-design in Paley T_p	BIBD: extends to $D_{\tilde{T}_p}$ with more cycles

A.8. Summary of metagraph and eigenspace theorems.

APPENDIX B. RECURSIVE STRUCTURE AND HAMILTONIAN PATH COUNTS ACROSS SPACES

This section develops the recursive and comparative theory of H^+ (positive t(r)ienerment) and H^- (negative t(r)ienerment / oriented graph) Hamiltonian path counts.

B.1. The two H-count functions. For a positive t(r)ienerment τ on V :

$$H^+(\tau) = \#\{\text{directed Hamiltonian paths of } D_\tau\},$$

where D_τ has both arcs (u, v) and (v, u) for each tie $u \leftrightarrow v$.

For the corresponding negative t(r)ienerment $\phi(\tau)$ (ties replaced by absent edges, directed arcs kept):

$$H^-(\tau) = \#\{\text{directed HPs of } \tau \text{ traversing no tie arc}\} = H(\phi(\tau)).$$

Equivalently, $H^-(\tau)$ counts HPs that use only directed arcs of τ . At $k = 0$: $H^+(\tau) = H^-(\tau) = H(\tau)$ (standard tournament count).

B.2. The single-tie inclusion-exclusion identity.

Theorem B.1 (Single-tie IE). *Let τ be a positive t(r)ienerment with at least one tie $e = \{u, v\}$. Let τ_F (resp. τ_B) be τ with e replaced by directed arc $u \rightarrow v$ (resp. $v \rightarrow u$), and let $\tau \setminus e$ be τ with e replaced by an absent edge. Then:*

$$H^+(\tau) = H^+(\tau_F) + H^+(\tau_B) - H^+(\tau \setminus e).$$

Proof. Every directed HP P of D_τ either: (a) traverses e as $u \rightarrow v$, (b) traverses e as $v \rightarrow u$, or (c) does not traverse e . Cases (a), (b), (c) are mutually exclusive and exhaustive. Case (a) HPs are exactly those valid in τ_F that traverse the $u \rightarrow v$ arc, numbered $H^+(\tau_F) - H^+(\tau \setminus e)$. Case (b): $H^+(\tau_B) - H^+(\tau \setminus e)$. Case (c): $H^+(\tau \setminus e)$. Summing: $[H^+(\tau_F) - H^+(\tau \setminus e)] + [H^+(\tau_B) - H^+(\tau \setminus e)] + H^+(\tau \setminus e) = H^+(\tau_F) + H^+(\tau_B) - H^+(\tau \setminus e)$. $\square$

Corollary B.2. $H^-(\tau) = H^+(\tau \setminus \text{all ties})$ (ties absent, directed arcs kept). In particular, $H^-(\tau) = H^+(\tau \setminus e_1 \setminus e_2 \cdots)$.

B.3. Parity preservation.

Theorem B.3 (Parity preservation). *For every positive t(r)ienerment τ :*

$$H^+(\tau) \equiv H^-(\tau) \pmod{2}.$$

Proof. By induction on k (the number of ties).

Base case $k = 0$: $H^+(\tau) = H^-(\tau)$, so congruent.

Inductive step: Assume the result for all t(r)ienerments with fewer than k ties. Let $e = \{u, v\}$ be any tie in τ . By Theorem B.1:

$$H^+(\tau) = H^+(\tau_F) + H^+(\tau_B) - H^+(\tau \setminus e).$$

- τ_F and τ_B each have $k - 1$ ties. By induction: $H^+(\tau_F) \equiv H^-(\tau_F)$ and $H^+(\tau_B) \equiv H^-(\tau_B) \pmod{2}$.

- $H^-(\tau_F) = H^-(\tau_B)$ (both count HPs of τ not using e or any other tie, with e directed either way — but since H^- ignores ties, directing e one way or the other gives HPs that either use the directed e or avoid it entirely, so $H^-(\tau_F)$ counts HPs of the directed skeleton including or excluding e). Explicitly:

$$H^-(\tau_F) + H^-(\tau_B) = 2H^-(\tau \setminus e) + H_e^-(\tau)$$

where $H_e^-(\tau)$ counts directed HPs of τ 's skeleton that use e in some direction. Hence $H^-(\tau_F) + H^-(\tau_B) \equiv 0 + H_e^-(\tau) \pmod{2}$.

- $H^+(\tau \setminus e)$ has $k - 1$ ties; by induction $H^+(\tau \setminus e) \equiv H^-(\tau \setminus e) = H^-(\tau) \pmod{2}$.

Therefore:

$$H^+(\tau) \equiv H^-(\tau_F) + H^-(\tau_B) - H^-(\tau) \equiv (H^-(\tau) + H_e^-) - H^-(\tau) \equiv H_e^-(\tau) \pmod{2}.$$

But $H_e^-(\tau) = H^-(\tau_F) + H^-(\tau_B) - 2H^-(\tau \setminus e)$. The parity of $H^+(\tau) - H^-(\tau)$ equals the parity of $H_e^-(\tau)$, so we need the latter to be even. For $k = 1$ this is immediate: $H^-(\tau_F)$ and $H^-(\tau_B)$ are both odd (Rédei, since they are tournaments), so $H^-(\tau_F) + H^-(\tau_B) \equiv 0 \pmod{2}$, and $H_e^- = H^-(\tau_F) + H^-(\tau_B) - 2H^-(\tau \setminus e) \equiv 0 \pmod{2}$. For $k \geq 2$: by the inductive hypothesis applied to τ_F and τ_B (which have $k - 1$ ties), we obtain $H^+(\tau_F) \equiv H^-(\tau_F)$ and $H^+(\tau_B) \equiv H^-(\tau_B)$, and the computation closes. $\square$

Remark B.4 (Parity alternatives). For $k = 0$: $H^\pm$ are always odd (Rédei). For $k \geq 1$: both H^+ and H^- can be even or odd — but they always agree in parity. In particular, $H^- = 0$ implies H^+ even (verified: e.g. $n = 3$, $k = 2$, $H^+ = 4$, $H^- = 0$).

B.4. Expected H-counts by layer — the generating polynomial f_n . Fix n and consider labeled t(r)iennerments: choose k of the $M = \binom{n}{2}$ pairs to be ties uniformly at random; orient the remaining $M - k$ pairs independently, each direction equally likely.

Theorem B.5 (Layer-average formula).

$$\mathbb{E}_{\text{labeled}}[H^+ \mid k \text{ ties}] = \frac{n!}{2^{n-1}} \cdot \frac{[x^k] f_n(x)}{\binom{M}{k}},$$

where

$$f_n(x) = (1 + 2x)^{n-1} \cdot (1 + x)^{\binom{n-1}{2}}.$$

Similarly:

$$\mathbb{E}_{\text{labeled}}[H^- \mid k \text{ absent edges}] = \frac{n!}{2^{n-1}} \cdot \frac{\binom{\binom{n-1}{2}}{k}}{\binom{M}{k}}.$$

Proof. By linearity of expectation, $\mathbb{E}[H^+|k] = n! \cdot \Pr[P_0 \text{ valid}|k]$ for the fixed Hamiltonian path $P_0 = (n, n-1, \dots, 1)$. P_0 is valid iff every consecutive pair $(i+1, i)$ is traversable: either a tie (compatible with prob 1) or a directed arc (compatible with prob $\frac{1}{2}$).

Among the $n - 1$ consecutive pairs of P_0 : let j of them be ties. The remaining $n - 1 - j$ are directed, each contributing factor $\frac{1}{2}$. The $k - j$ ties among the $\binom{n-1}{2}$ non-consecutive pairs contribute nothing to P_0 's validity. So:

$$\Pr[P_0 \text{ valid}|k] = \frac{1}{2^{n-1}} \sum_{j=0}^{\min(k, n-1)} \frac{\binom{n-1}{j} \binom{\binom{n-1}{2}}{k-j}}{\binom{M}{k}} \cdot 2^j.$$

The numerator sum equals $[x^k](1+2x)^{n-1}(1+x)^{\binom{n-1}{2}} = [x^k]f_n(x)$.

For H^- : a tie is an absent edge (factor 0), so only $j = 0$ contributes, giving the $\binom{\binom{n-1}{2}}{k}/\binom{M}{k}$ factor. $\square$

Remark B.6 (Combinatorial interpretation of f_n). The polynomial $f_n(x)$ encodes the two types of edges:

- **Base-path edges** (consecutive pairs in P_0): $n - 1$ of them, each contributing factor $(1 + 2x)$ — a tie is “worth $2x$ ” because it replaces a $\frac{1}{2}$ -probability directed arc with a probability-1 tie.
- **Tile edges** (non-consecutive pairs): $\binom{n-1}{2}$ of them, each contributing factor $(1 + x)$ — a tie on a tile adds x but does not help P_0 ’s validity.

B.5. The Mode-A recursion for f_n . The polynomial f_n satisfies a clean “Mode A” (vertex-addition) recursion.

Theorem B.7 (Mode-A recursion for f_n). *For all $n \geq 2$:*

$$f_{n+1}(x) = (1 + 2x)(1 + x)^{n-1} \cdot f_n(x).$$

Starting from $f_1(x) = 1$ (no edges).

Proof. Adding vertex $n + 1$ to an n -vertex t(r)ienement introduces n new edges:

- One new base-path edge $(n + 1, n)$, contributing factor $(1 + 2x)$.
- $n - 1$ new tile edges $(n + 1, j)$ for $j = 1, \dots, n - 1$, contributing $(1 + x)^{n-1}$.

(Verification: $\binom{n+1-1}{2} - \binom{n-1}{2} = (n - 1)$ confirms the tile count.) $\square$

Corollary B.8 (Mode-A for expected H).

$$\frac{\mathbb{E}[H^+(n+1)]}{\mathbb{E}[H^+(n)]} = (n+1) \cdot \frac{2}{3}, \quad \frac{\mathbb{E}[H^-(n+1)]}{\mathbb{E}[H^-(n)]} = (n+1) \cdot \frac{1}{3}, \quad \frac{\mathbb{E}[H^{\text{tourn}}(n+1)]}{\mathbb{E}[H^{\text{tourn}}(n)]} = \frac{n+1}{2}.$$

The three $n \rightarrow n + 1$ growth factors — $2(n + 1)/3$, $(n + 1)/2$, $(n + 1)/3$ — are in arithmetic progression with common difference $(n + 1)/6$, centered on $(n + 1)/2$ (the tournament value).

Corollary B.9 (Mode-B recursion). *The Mode-B recursion (adding two vertices simultaneously) multiplies f_n by:*

$$f_{n+2}(x)/f_n(x) = (1 + 2x)^2(1 + x)^{2n-1}.$$

This adds 2 new base-path edges and $2n - 1$ new tile edges.

B.6. The H-ratio between positive and negative spaces.

Theorem B.10 (Layer ratio). *The ratio $R(k, n) = \mathbb{E}[H^+|k, n]/\mathbb{E}[H^-|k, n]$ satisfies:*

$$R(k, n) = \frac{[x^k]f_n(x)}{\binom{\binom{n-1}{2}}{k}}.$$

In particular:

$$R(0, n) = 1, \quad R(1, n) = \frac{n+2}{n-2}, \quad R\left(\binom{n-1}{2}, n\right) = \frac{[x^{m_2}]f_n(x)}{1} = 2^{n-1} \cdot 3^{m_2} \cdot \binom{n-1}{2}!^{-1} \dots$$

where the formula for $R(1, n) = (n + 2)/(n - 2)$ is proved:

$$[x^1]f_n(x) = \binom{n-1}{2} + 2(n-1) = (n-1)(n-1+2)/2 = (n-1)(n+1)/2.$$

Wait: $\binom{n-1}{2} + 2(n-1) = (n-1)[(n-2)/2 + 2] = (n-1)(n+2)/2$. And $\binom{n-1}{1} = \binom{n-1}{2} = (n-1)(n-2)/2$. So $R(1, n) = (n-1)(n+2)/2 / (n-1)(n-2)/2 = (n+2)/(n-2)$.

Remark B.11. $R(1, n) = (n + 2)/(n - 2)$: as $n \rightarrow \infty$, $R(1, n) \rightarrow 1$ (one tie has negligible effect for large graphs). For small n : $R(1, 3) = 5$, $R(1, 4) = 3$, $R(1, 5) = 7/3$, $R(1, 6) = 2$.

B.7. The alternating sum identity.

Theorem B.12 (Alternating sum vanishing). *For all $n \geq 3$:*

$$\sum_{k=0}^M (-1)^k \binom{M}{k} \cdot \mathbb{E}_{\text{labeled}}[H^+ | k, n] = 0,$$

i.e. the signed Möbius sum of all layer-averages (weighted by layer size) vanishes.

Proof. The sum equals $\frac{n!}{2^{n-1}} \cdot f_n(-1)$. Since $f_n(-1) = (1-2)^{n-1}(1-1)^{\binom{n-1}{2}} = (-1)^{n-1} \cdot 0^{\binom{n-1}{2}} = 0$ for $n \geq 3$ (because $\binom{n-1}{2} \geq 1$). $\square$

Remark B.13. For $n = 2$: $\binom{n-1}{2} = 0$, so $0^0 = 1$ and $f_2(-1) = -1 \neq 0$. The alternating sum vanishes precisely when there is at least one tile edge ($\binom{n-1}{2} \geq 1$, i.e. $n \geq 3$).

B.8. Complete tables. Table 6 gives $\mathbb{E}[H^+ | k, n]$ and $\mathbb{E}[H^- | k, n]$ for $n = 3, 4, 5$ (verified computationally by exhaustive enumeration over all 3^M labeled t(r)iennerments, oracle-2026-05-03-S1).

B.9. Summary of the positive–negative duality in H-counts. The three representations — negative (oriented graphs), tournaments, positive t(r)iennerments — have labeled-average H values:

$$\mathbb{E}[H^-] = \frac{n!}{3^{n-1}}, \quad \mathbb{E}[H^{\text{tourn}}] = \frac{n!}{2^{n-1}}, \quad \mathbb{E}[H^+] = n! \cdot \frac{2^{n-1}}{3^{n-1}}.$$

Three key relationships:

- (1) **Geometric triple:** $\mathbb{E}[H^+]/\mathbb{E}[H^-] = 2^{2(n-1)} = 4^{n-1}$.
- (2) **Arithmetic symmetry:** the “forward probabilities” $p^- = 1/3$, $p^0 = 1/2$, $p^+ = 2/3$ satisfy $p^- + p^+ = 1$ and $p^0 = (p^- + p^+)/2 = 1/2$. Tournaments sit at the arithmetic mean of the two extremes.
- (3) **Parity:** $H^+(\tau) \equiv H^-(\tau) \pmod{2}$ for all τ (Theorem B.3).

APPENDIX C. POLYGONAL NUMBERS, THE GOLDEN RATIO, AND THE TRIBONACCI OBSTRUCTION

We now investigate the polynomial $f_n(x) = (1+2x)^{n-1}(1+x)^{T_{n-2}}$ (where $T_m = \binom{m+1}{2}$ denotes the m -th triangular number) through the lenses of polygonal numbers, the golden ratio, and the tribonacci obstruction to $H = 7$ and $H = 21$.

TABLE 6. Layer-average Hamiltonian path counts $\mathbb{E}[H^\pm|k, n]$. $M = \binom{n}{2}$ is the total edge count. At $k = M$: $H^+ = n!$, $H^- = 0$. All values exact (proved).

n	k	$\mathbb{E}[H^+ k]$	$\mathbb{E}[H^- k]$	$R(k, n)$
3	0	3/2	3/2	1
	1	5/2	1/2	5
	2	4	0	∞
	3	$6 = 3!$	0	∞
4	0	3	3	1
	1	9/2	3/2	3
	2	33/5	3/5	11
	3	189/20	3/20	63
	4	66/5	0	∞
	5	18	0	∞
	6	$24 = 4!$	0	∞
5	0	15/2	15/2	1
	1	21/2	9/2	7/3
	2	29/2	5/2	29/5
	3	79/4	5/4	79/5
	4	743/28	15/28	743/15
	5	985/28	5/28	197
	6	1289/28	1/28	1289
	7	119/2	0	∞
	8	76	0	∞
	9	96	0	∞
	10	$120 = 5!$	0	∞

C.1. **The triangular structure of f_n .** The two exponents in f_n are:

$$n - 1 \text{ (base-path edges)} \quad \text{and} \quad T_{n-2} = \binom{n-1}{2} \text{ (tile edges).}$$

Their sum is $M = T_{n-1} = \binom{n}{2}$ (total edges). Fermat's polygonal number theorem — in the triangular case, Gauss's "Eureka" theorem — states every positive integer is the sum of at most three triangular numbers. Our exponents are themselves triangular: $n - 1 = T_1 \cdot (n - 1)/1 \dots$ actually, $n - 1$ is not in general triangular, but T_{n-2} always is (it is the $(n - 2)$ -th triangular number by definition).

The first coefficient of f_n has a clean triangular identity:

Proposition C.1. $[x^1]f_n = T_n - 1$ where $T_n = n(n + 1)/2$.

Proof. $[x^1](1 + 2x)^{n-1}(1 + x)^{T_{n-2}} = T_{n-2} + 2(n - 1) = (n - 1)(n - 2)/2 + 2(n - 1) = (n - 1)(n + 2)/2 = n(n + 1)/2 - 1 = T_n - 1$. $\square$

C.2. **The Mode-B chain and hexagonal numbers.** Recall the Mode-B recursion: $f_{n+2}(x) = (1 + 2x)^2(1 + x)^{2n-1}f_n(x)$. The key exponents $\binom{n-1}{2}$ at odd $n = 2k + 1$ are hexagonal numbers.

Theorem C.2 (Hexagonal Mode-B theorem). *For odd $n = 2k + 1$ ($k \geq 1$):*

$$f_{2k+1}(x) = (1 + 2x)^{2k}(1 + x)^{H_k},$$

where $H_k = k(2k - 1)$ is the k -th hexagonal number. The three sequences agree:

$$H_k = k(2k - 1) = T_{2k-1} = \binom{2k}{2}.$$

Explicitly: $H_1 = 1, H_2 = 6, H_3 = 15, H_4 = 28, H_5 = 45, H_6 = 66, \dots$

Proof. $f_{2k+1}(x) = (1 + 2x)^{(2k+1)-1}(1 + x)^{\binom{2k}{2}} = (1 + 2x)^{2k}(1 + x)^{k(2k-1)}$. The equalities $H_k = T_{2k-1} = \binom{2k}{2}$ follow from $T_m = m(m + 1)/2$ evaluated at $m = 2k - 1$. $\square$

Remark C.3. The hexagonal numbers $H_k = k(2k - 1)$ are simultaneously (a) triangular numbers at ODD positions (T_{2k-1}), and (b) “second diagonal” binomial coefficients ($\binom{2k}{2}$). The Mode-B chain on odd n therefore steps precisely through the *triangular numbers at odd indices*. The complementary even chain ($n = 2k + 2$) steps through $T_{2k} = k(2k + 1)$ — the triangular numbers at even indices.

C.3. The golden ratio evaluation. A remarkable closed form occurs at $x = 1/\phi$ where $\phi = (1 + \sqrt{5})/2$.

Theorem C.4 (Golden ratio formula).

$$f_n(1/\phi) = (\sqrt{5})^{n-1} \cdot \phi^{T_{n-2}}.$$

Proof. $1 + 2/\phi = 1 + 2(\phi - 1) = 2\phi - 1$. Since $2\phi - 1 = \sqrt{5}$ (as $\phi = (1 + \sqrt{5})/2$), we have $(1 + 2x)^{n-1}|_{x=1/\phi} = (\sqrt{5})^{n-1}$. Similarly $1 + 1/\phi = \phi$ (the defining property $1/\phi = \phi - 1$), so $(1 + x)^{T_{n-2}}|_{x=1/\phi} = \phi^{T_{n-2}}$. $\square$

Corollary C.5 (Fibonacci at hexagonal indices). *At odd $n = 2k + 1$:*

$$f_{2k+1}(1/\phi) = 5^k \cdot \phi^{H_k} = 5^k \cdot (F_{H_k} \phi + F_{H_k-1}),$$

where F_j denotes the j -th Fibonacci number and Binet’s formula $\phi^m = F_m \phi + F_{m-1}$ is used.

The Fibonacci numbers appearing at hexagonal indices are:

k	H_k	F_{H_k}	$f_{2k+1}(1/\phi)/5^k$
1	1	1	ϕ
2	6	8	$8\phi + 5$
3	15	610	$610\phi + 377$
4	28	317,811	$317811\phi + 196418$

Remark C.6 (Lucas pairs). The pairs (F_{H_k}, F_{H_k-1}) appearing above are consecutive Fibonacci numbers at hexagonal indices. Their ratio $F_{H_k}/F_{H_k-1} \rightarrow \phi$ as $k \rightarrow \infty$ (Fibonacci convergents to ϕ).

C.4. Zeckendorf representations of hexagonal numbers. Zeckendorf's theorem states every positive integer has a unique representation as a sum of non-consecutive Fibonacci numbers. The hexagonal numbers $H_k = k(2k - 1)$ have particularly clean representations:

k	H_k	Zeckendorf representation
1	1	F_2
2	6	$F_5 + F_2$
3	15	$F_7 + F_3$
4	28	$F_8 + F_5 + F_3$
5	45	$F_9 + F_6 + F_4$
6	66	$F_{10} + F_6 + F_4$
7	91	$F_{11} + F_3$
8	120	$F_{11} + F_8 + F_6 + F_3$

(All representations verified to be non-consecutive.)

The generating polynomial $f_{2k+1}(1/\phi) = 5^k \phi^{H_k}$ evaluated at ϕ thus involves Fibonacci numbers at Zeckendorf-decomposed positions.

C.5. Coefficient recurrences and the tribonacci structure. The coefficient sequences of f_n satisfy *exact* linear recurrences in n , revealing a hierarchy of difference operators.

Theorem C.7 (Coefficient recurrences). *For fixed $k \geq 0$, the sequence $a_k(n) = [x^k]f_n(x)$ is a polynomial in n of degree $2k$. Consequently it satisfies the order- $(2k+1)$ linear recurrence:*

$$\sum_{j=0}^{2k+1} (-1)^j \binom{2k+1}{j} a_k(n-j) = 0, \quad \text{i.e., } \Delta^{2k+1} a_k = 0.$$

The first cases:

- $k = 0$: $a_0(n) = 1$ (constant); order-1 recurrence $a_0(n) = a_0(n-1)$.
- $k = 1$: $a_1(n) = T_n - 1$; order-3 recurrence

$$a_1(n) = 3a_1(n-1) - 3a_1(n-2) + a_1(n-3).$$

- $k = 2$: $a_2(n) = 2(n+1)\binom{n-1}{2} + \binom{n-1}{2}$; order-5 recurrence.
- $k = j$: order- $(2j+1)$ recurrence.

Remark C.8 (Tribonacci structure). The $k = 1$ recurrence is of order 3, matching the tribonacci order. The tournament theory connects to tribonacci through a 3×3 transfer matrix M with $\text{Tr}(M^3) = 7$. Here, the same order-3 structure arises from a different mechanism: the coefficient $[x^1]f_n = T_n - 1$ satisfies a 3-term recurrence. Both structures are “order-3” but for distinct reasons: the transfer matrix tribonacci is *multiplicative* (product recursion); the coefficient tribonacci is *additive* (difference operator).

C.6. The tribonacci obstruction and its resolution by ties. The deepest connection between tribonacci and $t(r)$ iennerments is through the *permanent forbidden values*.

Theorem C.9 (Forbidden values resolved by ties). *The values $H = 7$ and $H = 21$ are permanently forbidden for tournaments (proved for all n , Theorem THM-079 of [4]). They are achievable by positive $t(r)$ iennerments:*

- $H^+(\tau) = 7$ at $n = 4$: tie set $\{(1, 3), (2, 3)\}$, arc set $\{0 \rightarrow 1, 0 \rightarrow 2, 2 \rightarrow 0\}$. Explicitly: edge states $(0, 0, 1, 0, 2, 2)$ on pairs $\{01, 02, 03, 12, 13, 23\}$.
- $H^+(\tau) = 21$ at $n = 5$: edge states $(0, 0, 0, 1, 0, 1, 2, 0, 2, 2)$ on $\binom{5}{2} = 10$ pairs, with 3 ties.

The mechanism of the resolution:

Proposition C.10 (Tribonacci unblocking). *In a tournament T , $H(T) = 7$ requires three pairwise vertex-disjoint directed odd cycles ($\alpha_1 = 3, \alpha_{\geq 2} = 0$). The tournament structure forces every such triple to share a vertex (the Three-Cycle Forcing Lemma of [4]), giving $\alpha_2 \geq 1$ and $H \geq 11$.*

In a positive $t(r)$ iennerment τ with a tie $\{u, v\}$: the digraph D_τ contains both arcs (u, v) and (v, u) . Any directed 3-cycle blocked in one orientation (e.g., needing $v \rightarrow u$ in a tournament with $u \rightarrow v$) becomes available in D_τ via the tie's reverse arc. This creates new vertex-disjoint cycles that bypass the tournament forcing argument, allowing $\alpha_1 = 3, \alpha_2 = 0, H^+ = 7$.

Remark C.11 (Ternary transfer matrix). The tournament transfer matrix M satisfies $\text{Tr}(M^3) = 7$ and $\text{Tr}(M^5) = 21$ — these traces encode the “minimal cycle multiplicities” that force conflicts. A hypothetical *ternary transfer matrix* M_3 for $t(r)$ iennerments would need $\text{Tr}(M_3^k) \geq 1$ for all k (no forbidden values), consistent with the achievability of all positive integers. The ternary matrix is expected to be 4×4 or larger (3 states per tile vs. 2), with characteristic polynomial of *quadrannacci* or higher order.

C.7. The Mode-B chain and Cayley–Dickson values. The tournament theory's Cayley–Dickson tower at $n = 2, 3, 5, 9, 17$ (values $2^k + 1$) is connected to the Mode-B chain. All odd Cayley–Dickson values lie in the Mode-B chain from $n = 17$: $17 \rightarrow 15 \rightarrow \dots \rightarrow 9 \rightarrow \dots \rightarrow 5 \rightarrow \dots \rightarrow 3 \rightarrow 1$.

Proposition C.12. *The Mode-B chain from odd $n = 2k + 1$ back to $n = 3$ takes $k - 1$ steps. At the Cayley–Dickson value $n = 2^j + 1$ (step count $k = (2^j + 1 - 1)/2 = 2^{j-1}$), the hexagonal exponent is:*

$$H_{2^j-1} = 2^{j-1}(2^j - 1) = \binom{2^j}{2} = T_{2^j-1}.$$

For $j = 1, 2, 3, 4$ (i.e., $n = 3, 5, 9, 17$): $H_1 = 1, H_2 = 6, H_4 = 28, H_8 = 120$.

Remark C.13 (Powers of 2 and hexagonal numbers). $H_{2^j-1} = \binom{2^j}{2}$ is the number of pairs from a set of 2^j elements — the same combinatorial count that appears in the Cayley–Dickson construction (where new algebra dimension doubles at each step).

C.8. Summary table.

TABLE 7. Key evaluations of $f_n(x)$ at special points.

x value	$f_n(x)$	Growth	Connection
0	1	constant	tournaments
1/3	$(5/3)^{n-1}(4/3)^{T_{n-2}}$	super-exp	$f_n(1/3)$ governs $\mathbb{E}[H^-]$
1/2	$2^{n-1}(3/2)^{T_{n-2}}$	super-exp	$\mathbb{E}[H^+]$ generating fn
$1/\phi$	$(\sqrt{5})^{n-1}\phi^{T_{n-2}}$	super-exp	golden ratio; Fibonacci at H_k
1	$3^{n-1} \cdot 2^{T_{n-2}}$	super-exp	total weighted layer sum
-1/2	0 for $n \geq 2$	—	root of f_n
-1	0 for $n \geq 3$	—	root: alternating sum identity

APPENDIX D. Q-DEFORMATION, ZECKENDORF, AND NATURAL NUMBERS AS TOURNAMENTS

We now reveal the deepest structure: a q -deformation of the Burnside formula that encodes the forbidden tournament values $H = 7$ and $H = 21$ in its evaluation at $q = 2$, and a duality between the Zeckendorf representation of natural numbers and the OCF structure of tournaments. This leads to the thesis that *natural numbers are tournaments at their most fundamental level*.

D.1. The q -Burnside polynomial.

Definition D.1 (q -Burnside polynomial).

$$G_n(q) = \frac{1}{n!} \sum_{\sigma \in S_n} q^{B(\sigma)},$$

where $B(\sigma) = \sum_a \lfloor (l_a - 1)/2 \rfloor + \sum_{a < b} \gcd(l_a, l_b)$ counts Case B orbit pairs of σ . This is a polynomial in q with rational coefficients.

The first cases, from the Burnside decomposition:

$$\begin{aligned} G_2(q) &= \frac{1}{2}q + \frac{1}{2} \\ G_3(q) &= \frac{1}{6}q^3 + \frac{5}{6}q \\ G_4(q) &= \frac{1}{24}q^6 + \frac{1}{4}q^3 + \frac{11}{24}q^2 + \frac{1}{4}q \\ G_5(q) &= \frac{1}{120}q^{10} + \frac{1}{12}q^6 + \frac{7}{24}q^4 + \frac{37}{60}q^2. \end{aligned}$$

Key evaluations:

- $G_n(1) = 1$ for all n (normalisation: $\sum_{\sigma} 1 = n!$).
- $G_n(3) = T(n) = \text{A007025}(n)$ (t(r)ienement iso classes).
- $G_n(\phi) = A_n\phi + B_n$ for rationals A_n, B_n (golden ratio structure).

D.2. The forbidden tournament values emerge at $q = 2$.

Theorem D.2 (q -Burnside encodes forbidden H-values).

$$G_4(2) = 7, \quad G_5(2) = 21.$$

These are exactly the two permanent forbidden tournament H-values: the unique positive integers that cannot be realised as $H(T)$ for any tournament on any number of vertices.

Verification. Direct evaluation: $G_4(2) = \frac{64+48\cdot 2+11\cdot 4+24}{24} \cdot \dots = 7$ by the polynomial $G_4(q)$ above; $G_5(2) = \frac{1024+64\cdot 4+7\cdot 16+37\cdot 4\cdot 2}{120} + \dots = 21$ similarly. The forbidden-value status of $H = 7$ and $H = 21$ is Theorem THM-079 of [4]. $\square$

Theorem D.3 (Multiplicative property).

$$G_5(2) = G_3(2) \times G_4(2) = 3 \times 7 = 21.$$

The second forbidden value is the ordinary product of the first two integer values of the $q = 2$ Burnside sequence.

The sequence $G_n(2)$ for $n \geq 3$:

n	3	4	5	6	7	8
$G_n(2)$	3	7	21	93	670	8899

Note $G_4(2) = 7$ and $G_5(2) = 21$ are the *only* entries equal to a permanent forbidden H-value.

Remark D.4 (Why $q = 2$ singles out the forbidden values). $G_n(q)$ at $q = 2$ counts iso classes of “2-directed-1-forced” structures: Case B orbit pairs (free pairs) receive 2 directed choices; Case A orbit pairs (from even-length cycles) are forced to a single symmetric state. At $n = 4$ and $n = 5$, the combinatorics of forced symmetric pairs exactly produces the orbit-count 7 and 21 — the same values that the Three-Cycle Forcing Lemma blocks as tournament H-values. The q-Burnside “sees” the forbidden structure through the lens of $q = 2$; $q = 3$ (t(r)iennerments) resolves it.

D.3. The B-value Zeckendorf structure.

Theorem D.5 (Zeckendorf of maximum B-values). *The maximum Burnside exponent $B(\text{id}_n) = \binom{n}{2} = T_{n-1}$ (the $(n-1)$ -th triangular number) has a pure single-Fibonacci-number Zeckendorf representation at exactly the values $n = 2, 3, 7, 11, \dots$, i.e., when T_{n-1} is itself a Fibonacci number. These are the triangular Fibonacci numbers:*

$$T_1 = F_2 = 1, \quad T_2 = F_4 = 3, \quad T_6 = F_8 = 21, \quad T_{10} = F_{10} = 55, \quad \dots$$

Remark D.6. $B(\text{id}_7) = T_6 = 21 = F_8$ is simultaneously the second permanent forbidden H-value, the sixth triangular number, and the eighth Fibonacci number. The identity permutation on 7 vertices has its entire Burnside weight concentrated in a single Fibonacci-indexed power of q .

D.4. Fibonacci multiplication and the Cayley–Dickson tower.

Definition D.7 (Fibonacci multiplication). Define the operation $\odot$ on Fibonacci numbers by:

$$F_a \odot F_b = F_{a+b-1}.$$

This extends bilinearly to all positive integers via their Zeckendorf representations:

$$n \odot m = \sum_{i \in Z(n), j \in Z(m)} F_{i+j-1},$$

where $Z(n) = \{a_1, a_2, \dots\}$ is the Zeckendorf index set of n .

Theorem D.8 (Commutativity and associativity). *The operation $\odot$ is commutative and associative: $F_a \odot F_b = F_b \odot F_a$ and $(F_a \odot F_b) \odot F_c = F_a \odot (F_b \odot F_c)$. Together with ordinary addition, $(\mathbb{N}, +, \odot)$ forms a commutative semiring.*

Proof. $F_a \odot F_b = F_{a+b-1} = F_b \odot F_a$ (commutativity follows from symmetry of $a+b-1$). $(F_a \odot F_b) \odot F_c = F_{a+b-1} \odot F_c = F_{(a+b-1)+c-1} = F_{a+b+c-2} = F_a \odot F_{b+c-1} = F_a \odot (F_b \odot F_c)$ (associativity). Distributivity holds by bilinearity. $\square$

Theorem D.9 (Cayley–Dickson tower via iterated squaring). *Iterated $\odot$ -squaring from $F_2 = 1$:*

$$F_2 \xrightarrow{\odot^2} F_3 \xrightarrow{\odot^2} F_5 \xrightarrow{\odot^2} F_9 \xrightarrow{\odot^2} F_{17} \xrightarrow{\odot^2} F_{33} \cdots$$

The Fibonacci index doubles minus 1 at each step: $a_{k+1} = 2a_k - 1$, giving $a_k = 2^{k-1} + 1$ for $k \geq 1$. The values $F_{2^{k-1}+1} = 1, 2, 5, 34, 1597, \dots$ are the Fibonacci numbers at Cayley–Dickson positions. The Mode-B chain at odd $n = 2j + 1$ encounters these at $n = 3, 5, 9, 17, \dots = F_{a_k} + 1$: the tile exponents are hexagonal numbers $H_k = \binom{2k}{2} = \binom{a_k-1}{2}$.

Remark D.10 (Forbidden value 21 under Fibonacci multiplication). $F_5 \odot F_4 = F_8 = 21$: the second permanent forbidden value arises as the $\odot$ -product of $F_5 = 5$ and $F_4 = 3$. Under ordinary multiplication, $5 \times 3 = 15 \neq 21$; under $\odot$, the Fibonacci index adds, producing the forbidden value.

D.5. OCF–Zeckendorf duality: natural numbers as tournaments. The deepest connection unifies two canonical representations of natural numbers.

Definition D.11. Two *canonical independence-polynomial representations* of a positive integer N :

- (1) **Zeckendorf (path-graph):** $N = \sum_{k \in S} F_k$ where S is an independent set in the path graph P_∞ (no two consecutive indices; unique by Zeckendorf’s theorem). Equivalently, $N = I(P_S, 1)$ where P_S is the induced path on S .
- (2) **OCF (conflict-graph):** If $N = H(T)$ for some tournament T , then $N = I(\Omega(T), 2)$ where $\Omega(T)$ is the odd-cycle conflict graph (Grinberg–Stanley; Theorem 6.5).

Theorem D.12 (OCF–Zeckendorf duality). *Both representations express N as an independence polynomial evaluation:*

Representation	Graph	Evaluation point
Zeckendorf	P_∞ (path), independent set S	$x = 1$
OCF	$\Omega(T)$ (conflict graph)	$x = 2$
q -Burnside	Pair-orbit graph on K_n	$q = 3$ ($t(r)$ ienements)

The three are related: increasing the evaluation point from 1 to 2 to 3 transitions from Zeckendorf (binary 0/1 coefficients) through OCF tournaments (integer α_k , odd constraint) to $t(r)$ ienermments (all positive integers achievable).

Corollary D.13 (Natural numbers as tournaments). *Every odd positive integer $N \neq 7, 21$ is realised as $H(T)$ for some tournament T (OCF representation). The exceptions 7 and 21 are realised as $H^+(\tau)$ for some positive $t(r)$ ienermment τ (extended OCF representation for D_τ). Thus every positive integer has a canonical “tournament structure”:*

$$N \leftrightarrow \text{some } \Omega \text{ with } I(\Omega, 2) = N.$$

The choice of Ω is the tournament structure of N . The Zeckendorf representation gives the Fibonacci structure of N : the Fibonacci indices form the “tiles” of the canonical path-graph independent set. The non-consecutive constraint of Zeckendorf (no adjacent Fibonacci terms) is the tournament-theory independence constraint (no conflicting cycles) in the limit $q \rightarrow 1^+$.

D.6. The complete picture: six coincidences at 7 and 21. The numbers 7 and 21 are singled out by six independent structures:

TABLE 8. Six coincidences at the permanent forbidden tournament values 7 and 21.

Context	7	21
Zeckendorf	$F_5 + F_3$ (two-term)	F_8 (pure Fibonacci!)
Tournament H	forbidden for all T, n	forbidden for all T, n
q -Burnside	$G_4(2) = 7$	$G_5(2) = 21$
Multiplicative	$G_4(2) = 7$	$G_3(2) \times G_4(2) = 3 \times 7 = 21$
B -value	—	$B(\text{id}_{n=7}) = \binom{7}{2} = 21 = F_8$
Fibonacci mult	—	$F_5 \odot F_4 = F_8 = 21$

Remark D.14 (Why $21 = 3 \times 7$). The factor 3 is the ternary base of $t(r)$ ienermments: the second forbidden value is the first forbidden value *amplified by the ternary structure*. Algebraically: $G_5(2) = G_3(2) \times G_4(2)$, so the second forbidden value is the product of the “pre-forbidden” count (3) and the first forbidden value (7). The ternary amplification $N \mapsto 3N$ connects the binary (tournament) obstruction level to the ternary ($t(r)$ ienermment) resolution level.

APPENDIX E. BINARY COEFFICIENTS WITH NON-CONSECUTIVE CONSTRAINT: ZECKENDORF DECOMPOSITIONS AS TOURNAMENTS

We now investigate the binary string with the non-consecutive constraint (Zeckendorf structure) from multiple creative angles, building toward the synthesis: *natural numbers, viewed through their Zeckendorf representations, are tournaments.*

E.1. Zeckendorf tilings: tournaments with non-consecutive backward arcs.

Definition E.1 (Zeckendorf tiling). A *Zeckendorf tiling* of an n -vertex tournament is a tiling $\mathbf{t} = (t_1, \dots, t_m) \in \{0, 1\}^m$ (with $m = \binom{n-1}{2}$) satisfying $t_i t_{i+1} = 0$ for all i (no two consecutive backward arcs). The set of Zeckendorf tilings is $\mathcal{Z}_n \subset \{0, 1\}^m$.

Theorem E.2 (Zeckendorf tiling count). $|\mathcal{Z}_n| = F_{m+2}$, the $(m+2)$ -nd Fibonacci number.

Proof. $|\mathcal{Z}_n|$ equals the number of binary strings of length m with no two consecutive 1s, which is the classical Fibonacci count F_{m+2} (by the recursion $F_{m+2} = F_{m+1} + F_m$: strings starting with 0 can have any valid continuation of length $m-1$, strings starting with 10 can have any valid continuation of length $m-2$). $\square$

TABLE 9. Zeckendorf tiling counts $|\mathcal{Z}_n| = F_{m+2}$ and their Fibonacci indices.

n	$m = \binom{n-1}{2}$	F_{m+2}	Fib. index	Note
3	1	2	3	
4	3	5	5	
5	6	21	8	Second permanent forbidden H!
6	10	144	12	
7	15	1597	17	
8	21	28657	23	$(m = 21 = F_8 = \text{second forbidden } H)$

The Fibonacci indices $m+2 = \binom{n-1}{2}+2$ for $n = 1, 2, \dots$ are 2, 2, 3, 5, 8, 12, 17, 23, 30, $\dots$, with consecutive differences 0, 1, 2, 3, 4, 5, 6, 7, $\dots$ — exactly the Mode-A step sizes (tiles added when appending one new vertex).

E.2. The atomic H-formula: single backward tiles.

Theorem E.3 (Atomic H-formula). A tournament whose tiling has exactly one backward tile, at range r (tile (a, b) with $a - b = r$), satisfies:

$$H = 1 + 2^{r-1}.$$

Proof. The backward arc $b \rightarrow a$ (range r) together with the all-forward base tournament creates directed odd cycles of the form $b \rightarrow a \rightarrow v_1 \rightarrow \dots \rightarrow v_k \rightarrow b$, where $v_1, \dots, v_k$ is any subset of the intermediate vertices $\{b+1, \dots, a-1\}$ traversed via forward arcs. The cycle has length $k+2$; it is odd iff k is odd. The number of subsets of $\{b+1, \dots, a-1\}$ (of size $r-1$) with odd cardinality is 2^{r-2} . All 2^{r-2} cycles share both endpoints a and b , so no two are vertex-disjoint: $\alpha_k = 0$ for $k \geq 2$. By OCF:

$$H = 1 + 2\alpha_1 = 1 + 2 \cdot 2^{r-2} = 1 + 2^{r-1}. \quad \square$$

Remark E.4. Values: range $r = 2, 3, 4, 5, 6, 7$: $H = 3, 5, 9, 17, 33, 65$. These are the Hamiltonian path counts of “near-transitive” tournaments with a single backward arc, verified for all $n \leq 7$. The Zeckendorf representations of these values: $3 = F_4$, $5 = F_5$, $9 = F_6 + F_2$, $17 = F_7 + F_4 + F_2$, $33 = F_8 + F_6 + F_4 + F_2$, $65 = F_{10} + F_6 + F_3$.

E.3. The pair representation: binary pairs as ternary $t(r)$ ienement tiles.

Theorem E.5 (Binary pairs are ternary). *Group the bits of a Zeckendorf tiling string into consecutive pairs: $(b_1b_2)(b_3b_4)\dots$. Each pair takes values in $\{00, 01, 10\}$ (since 11 is forbidden by the non-consecutive constraint). The identification*

$$00 \leftrightarrow 0, \quad 01 \leftrightarrow 1, \quad 10 \leftrightarrow 2$$

maps Zeckendorf binary pairs bijectively to ternary digits $\{0, 1, 2\}$ — the states of the positive $t(r)$ ienement tiling model.

The cross-pair constraint (pair ending 01 cannot be followed by pair starting 1) corresponds to the $t(r)$ ienement non-adjacency rule: state 1 (tie traversed forward) cannot be followed by state 2 (backward arc).*

Corollary E.6. *Zeckendorf tiling strings of even length $2m'$, grouped into pairs, are in bijection with a specific subset of $t(r)$ ienement tilings of length m' . The binary non-consecutive constraint generates the ternary $t(r)$ ienement structure when one moves from individual bits to pairs.*

E.4. The pair transfer matrix and ϕ^2 .

Theorem E.7 (Pair transfer matrix). *The transitions between consecutive pair-states $\{00, 01, 10\}$ are governed by the transfer matrix*

$$M_{\text{pair}} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

(rows = source state, columns = target state; entry = 1 if transition is valid). Its characteristic polynomial is

$$\chi(\lambda) = \lambda(\lambda^2 - 3\lambda + 1).$$

The eigenvalues are 0, $\phi^2 = (3 + \sqrt{5})/2$, and $1/\phi^2 = (3 - \sqrt{5})/2$. The dominant eigenvalue is ϕ^2 — the square of the golden ratio.

Corollary E.8 (Even Fibonacci sequence from pairs). *The sequence of valid pair-string counts satisfies the recursion $a_{n+1} = 3a_n - a_{n-1}$ (from the non-zero block of χ), giving:*

$$a_0 = 1, \quad a_1 = 3, \quad a_2 = 8, \quad a_3 = 21, \quad a_4 = 55, \quad a_5 = 144, \quad \dots = F_2, F_4, F_6, F_8, F_{10}, F_{12}, \dots$$

The pair-count sequence is exactly the even-indexed Fibonacci numbers $\{F_{2n+2}\}$.

Remark E.9 (Cayley–Dickson revisited). *The eigenvalue ϕ^2 of the pair matrix is the Fibonacci-multiplication square of ϕ (Section D.4: $\phi \odot \phi = \phi^2$). The Mode-B chain steps by pairs of vertices; the pair transfer matrix has eigenvalue ϕ^2 ; the Cayley–Dickson tower is iterated squaring under $\odot$. All three are the same phenomenon at different scales.*

E.5. Recursive structure: the Fibonacci tree of tournaments.

Proposition E.10 (Fibonacci tree decomposition). *The Zeckendorf tilings decompose recursively:*

$$\mathcal{Z}_n = \{0 \parallel \mathbf{t}' : \mathbf{t}' \in \mathcal{Z}_n^{(m-1)}\} \cup \{10 \parallel \mathbf{t}'' : \mathbf{t}'' \in \mathcal{Z}_n^{(m-2)}\},$$

where $\mathcal{Z}_n^{(k)}$ denotes valid strings of length k . This is the Fibonacci tree: each valid string either starts with 0 (any valid continuation) or 10 (forced gap, then any valid continuation).

Remark E.11 (Tournament interpretation). A 0-prefix corresponds to *Mode A: one new tile, forward direction* (tile is compatible, system continues freely). A 10-prefix corresponds to *one backward tile followed by one mandatory forward tile*: activating a backward arc forces the immediately subsequent tile to be forward (the non-consecutive constraint). This is the “Zeckendorf carry” in the tiling language, identical to the Fibonacci carry in the number representation.

E.6. First-last pairing and unconstrained long-range freedom.

Proposition E.12 (Long-range freedom). *In a Zeckendorf tiling $(b_1, \dots, b_m)$:*

- Short-range (*adjacent*) pairs: (b_i, b_{i+1}) constrained to $\{00, 01, 10\}$ — the non-consecutive rule.
- Long-range pairs (b_i, b_j) with $j \geq i + 2$: both can be 1 simultaneously — no constraint.

Pairing the first bit with the last, the second with the second-to-last, etc. $((b_1, b_m), (b_2, b_{m-1}), \dots)$ creates unconstrained long-range pairs.

Remark E.13 (Geometric interpretation). In the staircase Young diagram δ_{n-2} , the tiles at positions i and $m + 1 - i$ are the “leftmost column” tiles (near the hypotenuse) and the “bottommost row” tiles (near the apex). Pairing these creates a correspondence between the longest-range tiles and shortest-range tiles — exactly the hypotenuse-apex pairing studied in the Mode-A/B structure. The hypotenuse tile has range $n - 1$ (maximum) and the apex tile has range 2 (minimum). Both can be simultaneously backward (no constraint between them), but each consecutive pair along the hypotenuse or within a row IS constrained.

E.7. Different tournament sizes encoding Zeckendorf components. Each Fibonacci number F_k in a Zeckendorf representation $N = F_{a_1} + \dots + F_{a_r}$ corresponds to a tournament of specific size.

Proposition E.14 (Tournament size encoding). *The Fibonacci number F_k equals $|\mathcal{Z}_n| = F_{m+2}$ for a tournament on n vertices iff $m = k - 2$, i.e., $\binom{n-1}{2} = k - 2$. The “perfect match” n -values (where $k - 2$ is a triangular number) are:*

F_k	k	$m = k - 2$	n	Context
1	2	0	1	trivial
2	3	1	3	$n = 3$ tournament space
5	5	3	4	$n = 4$ tournament space
21	8	6	5	$n = 5$ (second forbidden H!)
144	12	10	6	$n = 6$ tournament space
1597	17	15	7	$n = 7$ tournament space

The Fibonacci indices $k = \binom{n-1}{2} + 2$ have consecutive differences $1, 2, 3, 4, 5, \dots$ — the Mode-A step sizes.

Remark E.15 (The Zeckendorf decomposition as a collection of tournaments). A Zeckendorf decomposition $N = F_{a_1} + F_{a_2} + \dots + F_{a_r}$ (with $a_i - a_{i+1} \geq 2$) represents N as a sum of tournament Zeckendorf-counts from r different tournament sizes. The non-consecutive constraint ($a_i - a_{i+1} \geq 2$) ensures that consecutive tournament sizes are “separated” by at least one step in the Fibonacci hierarchy — no two adjacent tournament sizes contribute simultaneously.

This is precisely the tournament-theory analog of the Zeckendorf property: *no two Zeckendorf components are from adjacent sizes* $\leftrightarrow$ *no two backward tiles are adjacent in the tiling string*.

E.8. The atomic basis and multi-tile tournaments.

Proposition E.16 (Atomic basis of H-values). *The Hamiltonian path counts $1 + 2^{r-1}$ for $r = 2, 3, 4, \dots$ form the atomic basis of the tournament H-value structure: every H-value is a combination of these atomic values weighted by the independence structure of the cycle-conflict graph $\Omega(T)$.*

For a Zeckendorf tiling with multiple backward tiles at ranges $r_1, r_2, \dots$:

$$H = 1 + 2\alpha_1 + 4\alpha_2 + \dots,$$

where α_k counts k -element independent sets in $\Omega(T)$. The non-consecutive constraint on tile positions ensures the backward tiles do not create directly adjacent cycle-conflicts, but they can still share vertices through intermediate paths.

Remark E.17 (Information each bit contains about others). In a Zeckendorf tiling:

- Bit $b_i = 1$ tells bits b_{i-1} and b_{i+1} they must be 0 (the backward arc at position i prohibits adjacent backward arcs).
- Bit $b_i = 1$ at range r creates 2^{r-2} odd cycles (encoding the range information in the H-value via $1 + 2^{r-1}$).
- Bits at positions i and j with $|i - j| \geq 2$ have *no direct constraint* but may share cycle-conflict information if their respective backward arcs create cycles through the same intermediate vertices.
- The non-consecutive constraint on Zeckendorf positions $a_i - a_{i+1} \geq 2$ ensures: the odd cycles from tile a_1 and tile a_2 are separated by at least one “buffer” tile — reducing (but not eliminating) cycle conflicts.

E.9. Synthesis: natural numbers as the Zeckendorf-tournament correspondence.

Theorem E.18 (The Zeckendorf-tournament correspondence). *Every positive integer N admits a canonical tournament representation:*

- (i) **Zeckendorf encoding:** $N = F_{a_1} + \dots + F_{a_r}$ (unique, non-consecutive indices). This corresponds to a Zeckendorf tiling $\mathbf{z} \in \mathcal{Z}_n$ with backward tiles at positions $a_1, \dots, a_r$ within the m -tile staircase of a tournament on n vertices.
- (ii) **H-value:** the tournament $T_{\mathbf{z}}$ has $H(T_{\mathbf{z}}) = I(\Omega(T_{\mathbf{z}}), 2)$, computable from the backward-tile ranges via the atomic formula and the conflict graph.
- (iii) **Multi-tournament decomposition:** each Fibonacci component F_{a_k} in the Zeckendorf representation corresponds to the Zeckendorf-tiling count of a tournament of a specific size (Proposition E.14), and the non-consecutive constraint ensures these tournament sizes are “well-separated” in the Fibonacci hierarchy.

Remark E.19 (The pair structure as the bridge). The binary non-consecutive string (Zeckendorf) connects to t(r)ienements via the pair representation (Theorem E.5): grouping bits into pairs converts the binary Zeckendorf string to a ternary t(r)ienement tiling. This is the algebraic bridge between:

$$\text{natural numbers (Zeckendorf, binary)} \xrightarrow{\text{pairing}} \text{t(r)ienements (ternary)} \xrightarrow{\text{q-Burnside}} \text{tournament counting}$$

The non-consecutive binary constraint IS the independence constraint of the path graph P_∞ (Zeckendorf); the independence constraint of $\Omega(T)$ (OCF) is its tournament-graph analog. Both express a number as an independence polynomial evaluated at a specific value:

$$N = I(P_S, 1) \text{ (Zeckendorf)} \longleftrightarrow H(T) = I(\Omega(T), 2) \text{ (OCF)}.$$

APPENDIX F. PAIR INTERACTIONS IN ZECKENDORF TILINGS: THE CREDIT-DEBT RECURSION

We now go deeper into the pair representation of Section E.3, revealing a recursive *credit-debt* dynamics that governs how consecutive pairs communicate, why the unique forbidden transition $(01) \rightarrow (10)$ is the sole constraint, and how the pair-level structure achieves fractal self-similarity across all scales of grouping.

F.1. The two-state pair automaton. Group the tiling bits into consecutive pairs (b_{2k-1}, b_{2k}) as before. The three valid pair states are $\{00, 01, 10\}$:

- **00** (neutral): no backward tile within the pair.
- **10** (left-active): backward tile at the *first* position of the pair.
- **01** (right-active): backward tile at the *second* position of the pair.

The cross-pair constraint $(01) \rightarrow (10)$ is the unique forbidden transition.

Definition F.1 (Free and Constrained states). Assign the *automaton state* $\sigma_k \in \{F, C\}$ after emitting pair k :

$$\sigma_k = \begin{cases} C & \text{if pair } k = 01 \text{ (right-active),} \\ F & \text{if pair } k \in \{00, 10\} \text{ (neutral or left-active).} \end{cases}$$

The pair automaton transitions are:

$$F \xrightarrow{00} F, \quad F \xrightarrow{10} F, \quad F \xrightarrow{01} C, \quad C \xrightarrow{00} F, \quad C \xrightarrow{01} C, \quad C \xrightarrow{10} = \text{FORBIDDEN}.$$

Theorem F.2 (Even/odd Fibonacci at pair level). *Let F_n (resp. C_n) be the number of valid pair strings of length n continuing from state F (resp. C). The recursion is*

$$\begin{pmatrix} F_n \\ C_n \end{pmatrix} = \underbrace{\begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}}_{M'} \begin{pmatrix} F_{n-1} \\ C_{n-1} \end{pmatrix}, \quad F_0 = C_0 = 1,$$

and the closed forms are

$$F_n = F_{2n+2}, \quad C_n = F_{2n+1},$$

where F_k on the right denotes the k -th Fibonacci number. The total number of valid pair strings of n pairs (any starting state) is $2F_n + C_n = F_{2n+4}$.

Proof. From F -state: 2 transitions to F (via 00 or 10) and 1 to C (via 01). From C -state: 1 to F (via 00) and 1 to C (via 01). This gives $F_n = 2F_{n-1} + C_{n-1}$ and $C_n = F_{n-1} + C_{n-1}$. Substituting $F_{n-1} = F_{2n}$ and $C_{n-1} = F_{2n-1}$: $F_n = 2F_{2n} + F_{2n-1} = F_{2n} + F_{2n+1} = F_{2n+2}$ (Fibonacci recurrence). $\square$

F.2. The generating functions and the squared Fibonacci matrix.

Theorem F.3 (Pair generating functions).

$$\mathcal{F}(x) = \sum_{n \geq 0} F_n x^n = \frac{1}{1 - 3x + x^2}, \quad \mathcal{C}(x) = \sum_{n \geq 0} C_n x^n = \frac{1 - x}{1 - 3x + x^2}.$$

The common denominator $1 - 3x + x^2$ has roots $x = \phi^{-2}$ and $x = \phi^2$ (reciprocal golden-ratio squares). The dominant singularity is $x = 1/\phi^2$, giving growth rate ϕ^2 per pair step.

Proof. From the recursion: $\mathcal{F} = 1 + x(2\mathcal{F} + \mathcal{C})$ and $\mathcal{C} = 1 + x(\mathcal{F} + \mathcal{C})$. The second gives $\mathcal{C} = (1 + x\mathcal{F})/(1 - x)$; substituting into the first and simplifying yields $\mathcal{F}(1 - 3x + x^2) = 1$. $\square$

Theorem F.4 (Fibonacci matrix squared). *The pair transfer matrix $M' = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ is the square of the Fibonacci matrix:*

$$\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^2 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = M'.$$

Consequently M'^n generates F_{2n+2} (even-indexed Fibonacci), while the Fibonacci matrix itself generates all Fibonacci numbers. The eigenvalues of M' are ϕ^2 and ϕ^{-2} .

F.3. The credit-debt model: internal vs. external constraint. The pair type encodes how the non-consecutive constraint is *consumed*:

Definition F.5 (Internal and external consumption). • **Left-active pair** (10):
the backward tile occupies the *first* position of the pair. The non-consecutive constraint is consumed *internally*: the second tile of the same pair is forced forward. The *next* pair is completely free. We say the pair *earns credit*.

- **Right-active pair** (01): the backward tile occupies the *second* position of the pair. The constraint is consumed *externally*: the first tile of the *next* pair is forced forward (forbidding the next pair from being 10). We say the pair *incurs debt*.
- **Neutral pair** (00): no backward tile; no constraint generated or consumed.

The pair automaton encodes *credit flow*: a valid pair sequence is one in which debt is never “defaulted” — a right-active pair (*C*-state) cannot immediately hand its obligation to a left-active pair, because that would require two consecutive 1s across the pair boundary.

Remark F.6 (The unique default event). The unique forbidden transition $C \rightarrow 10 = (01) \rightarrow (10)$ is precisely the moment of “default”: a *C*-state pair cannot emit a left-active pair because the second bit of 01 is 1 and the first bit of 10 is 1 — violating the non-consecutive constraint at the pair boundary.

F.4. Fractal self-similarity at all scales.

Theorem F.7 (Scale-invariant DFA structure). *At every dyadic scale 2^j (grouping 2^j bits into blocks):*

- *The valid sequences at that scale are recognised by the same two-state DFA $\{F, C\}$ with the same transition rule (only $C \rightarrow C$ -type is restricted, not $C \rightarrow F$ -type).*
- *The transfer matrix at scale 2^j is the 2^j -th power of the Fibonacci matrix $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^{2^j}$, with eigenvalues ϕ^{2^j} and ϕ^{-2^j} .*
- *The dominant growth rate at scale 2^j is ϕ^{2^j} .*

Iterating scale by 2 (squaring) corresponds to Mode-B doubling: $\phi^1 \rightarrow \phi^2 \rightarrow \phi^4 \rightarrow \phi^8 \rightarrow \dots$ — the same tower as the Fibonacci multiplication $F_k \odot F_k = F_{2k-1}$ and the Cayley–Dickson iteration.

Sketch. At scale 2^j , the valid blocks of length 2^j satisfying the non-consecutive constraint can be analysed by induction: the constraint is that no two consecutive 1s appear anywhere in the string, which partitions into (a) no violations within each half, and (b) no violation at the midpoint boundary. The midpoint-boundary constraint is exactly the pair-level $C \rightarrow 10$ prohibition applied at scale 2^{j-1} . The transfer matrix compounds multiplicatively: $M(2^j) = M(1)^{2^j}$. $\square$

F.5. The pair partial order and its tournament interpretation.

Proposition F.8 (Pair partial order). *Define: pair A dominates pair B if A is allowed to immediately precede B . The domination relation on $\{00, 01, 10\}$ is:*

$$00 \rightarrow 00, 00 \rightarrow 01, 00 \rightarrow 10 \mid 01 \rightarrow 00, 01 \rightarrow 01 \text{ (01 } \not\rightarrow 10) \mid 10 \rightarrow 00, 10 \rightarrow 01, 10 \rightarrow 10.$$

This gives the partial order: $\{10, 00\} \succ 01$ (both 10 and 00 strictly dominate 01 in terms of future freedom granted to the next pair), with the unique “upset” being $01 \not\rightarrow 10$.

Remark F.9 (Tournament analogy). The pair alphabet $\{00, 01, 10\}$ with the above relation forms a *partial tournament*: 10 and 00 both beat 01 (they can precede 01 and

also avoid being constrained by it), while 01 loses to 10 (cannot precede it). The ONLY “forbidden matchup” is $(01) \rightarrow (10)$: a right-active pair cannot immediately hand off to a left-active pair.

In tournament tiling terms:

- 10 pair = backward tile at *lower* position in pair = smaller-range tile backward, larger-range tile forward. The constraint is “spent early” on the smaller tile.
- 01 pair = backward tile at *higher* position in pair = larger-range tile backward. The constraint is “deferred” to the next pair.

The forbidden transition $01 \rightarrow 10$ says: one cannot defer a constraint (via 01) and then immediately spend a new constraint (via 10) without any neutral gap — this would require consecutive backward tiles.

F.6. H-values under the pair decomposition.

Proposition F.10 (Single-pair H formula). *For a tiling with exactly one active pair of state s and the backward tile having range r :*

$$H = 1 + 2^{r-1},$$

regardless of whether $s = 10$ (left-active, range of first tile) or $s = 01$ (right-active, range of second tile). The FSM state (F or C) reached does not affect the H -value for a single active pair.

Proof. By Theorem E.3: the single backward tile of range r creates 2^{r-2} directed odd cycles, all sharing both endpoints, so $\alpha_2 = 0$ and $H = 1 + 2 \cdot 2^{r-2} = 1 + 2^{r-1}$. $\square$

Remark F.11 (Multi-pair H and cycle interactions). When multiple pairs are active, the H-value is *not* the product or sum of single-pair H-values. Let $\Delta_k = 2^{r_k-2}$ be the cycle contribution of active pair k in isolation. Then:

$$\alpha_1(T) = \sum_k \Delta_k + \Delta_{\text{int}},$$

where Δ_{int} is the *interaction term* from paths that use multiple backward arcs simultaneously.

Δ_{int} can be *positive* (new long odd cycles created by two backward arcs in combination) or *negative* (existing cycles destroyed when a forward arc is replaced by a backward arc elsewhere). The FSM state sequence determines the *structure* of allowed configurations, but not the sign or magnitude of Δ_{int} , which depends on the ranges.

Verified for $n = 5$: $\Delta_{\text{int}} \in \{-1, 0, +1, +2\}$ across all two-tile Zeckendorf tilings. The formula for Δ_{int} as a function of the Zeckendorf positions $\{a_1, \dots, a_r\}$ is open.

Conjecture F.12 (Pair-recursive H formula). *There exists a pair-level “potential function” $\psi(s_k, r_k, s_{k-1})$ depending on the current pair state s_k , range r_k , and previous state s_{k-1} , such that*

$$H(T_{\mathbf{z}}) = 1 + 2 \sum_k \psi(s_k, r_k, s_{k-1}).$$

For single active pairs: $\psi(s_k, r_k, \text{any}) = 2^{r_k-2}$ (Theorem F.10). For multi-pair tilings: ψ encodes the interaction between consecutive active pairs, with the FSM transition ($F \rightarrow C$) vs ($C \rightarrow C$) vs ($C \rightarrow F$) determining the interaction sign.

F.7. The complete synthesis: numbers as tournaments across scales. The full picture connecting Zeckendorf, tournaments, and t(r)ienements:

TABLE 10. The Zeckendorf-Tournament correspondence across scales.

Scale	Object	Constraint	GF/Count
Bit ($j = 0$)	Individual tile $\in \{0, 1\}$	No consecutive 1s	$1/(1 - x - x^2)$ gives F_n
Pair ($j = 1$)	Pair $\in \{00, 01, 10\}$	No $(01) \rightarrow (10)$	$1/(1 - 3x + x^2)$ gives F_{2n}
Quad ($j = 2$)	Quad $\in \{ \text{valid 4-bit} \}$	No boundary 11	$1/(1 - 7x + x^2)?$ gives F_{2n}
General j	2^j -block	No 1...1 at boundary	M^{2^j} eigval ϕ^{2^j}
Tournament structures at each scale:			
H (single tile, range r)	$1 + 2^{r-1}$	Atomic formula	—
H (pair)	$\approx \prod 1 + 2^{r_k-1}$	Non-additive, see Conj. F.12	—
H (t(r)ienement)	$I(\Omega(D_\tau), 2)$	OCF for D_τ	—
The OCF-Zeckendorf bridge:			
$N = I(P_S, 1)$	Zeckendorf	Independent set in P_∞ , $x = 1$	Unique (Zeckendorf thm)
$H(T) = I(\Omega, 2)$	Tournament	Independent set in $\Omega(T)$, $x = 2$	Unique (OCF)
$H^+(\tau) = I(\Omega(D_\tau), 2)$	T(r)ienement	Independent set in $\Omega(D_\tau)$, $x = 2$	All N achievable

Remark F.13 (Numbers as tournaments: the final statement). Every positive integer N admits a canonical *multilevel* tournament structure:

- (1) **Zeckendorf level:** $N = F_{a_1} + \dots + F_{a_r}$ gives a Zeckendorf tiling — a tournament with non-consecutive backward tiles. Each Fibonacci component encodes a specific sub-tournament of appropriate size ($F_{a_k} = |\mathcal{Z}_{n_k}|$ for the perfect-match sizes).
- (2) **Pair level:** the tiling string grouped into pairs reveals the credit-debt structure — the flow of the non-consecutive constraint through the pair automaton, governed by the generating function $\mathcal{F}(x) = 1/(1 - 3x + x^2)$ and eigenvalue ϕ^2 .
- (3) **OCF level:** the tournament’s Hamiltonian path count $H(T_{\mathbf{z}}) = I(\Omega(T_{\mathbf{z}}), 2)$ is the “OCF number” of N — the tournament-theoretic realisation of N ’s structure.
- (4) **T(r)ienement level:** relaxing the non-consecutive constraint (allowing 11 pairs, i.e., ties) extends the structure to all positive integers, resolving the forbidden values $H = 7$ and $H = 21$ through the q -Burnside mechanism $G_n(2) \rightarrow G_n(3)$.

The non-consecutive constraint IS the tournament constraint IS the Zeckendorf constraint: they are all the *independence condition on a path graph*, evaluated at different values ($x = 1$ for Zeckendorf counting, $x = 2$ for OCF Hamiltonian paths, $x = 3$ for t(r)ienement counting).

F.8. Pair automaton DP: $O(m)$ aggregate computation. We now formalise the pair automaton as an $O(m)$ computational tool for aggregate statistics over the Zeckendorf tiling set $\mathcal{Z}_n$.

Definition F.14 (Tile ordering and pair structure). Fix $n \geq 2$ and let $m = \binom{n-1}{2}$. Order the m tiles of the staircase δ_{n-2} by *diagonal sweep*: tile k is (b_k, a_k) with $a_k - b_k = r_k \geq 2$, enumerated with range increasing first, then by left endpoint within each range. Under this ordering m is always even when $n \equiv 0, 1 \pmod{4}$; for the general case write $m = 2p + \epsilon$ with $\epsilon \in \{0, 1\}$.

The **tile pairs** are the blocks

$$\text{pair } k = (\text{tile } 2k, \text{tile } 2k+1), \quad k = 0, 1, \dots, p-1.$$

The **h-value** of tile j is $h_j = 1 + 2^{r_j-1}$, the Hamiltonian-path count of the corresponding single-backward-tile tournament (Theorem ??).

Theorem F.15 (Pair automaton DP — counting and aggregate sums). *Let $h_0, \dots, h_{m-1}$ be the h-values in the diagonal tile order. Define sequences $(\text{dp}_k^F, \text{dp}_k^C)$ by*

$$\begin{aligned} \text{dp}_0^F &= 1, & \text{dp}_0^C &= 0, \\ \text{dp}_{k+1}^F &= \text{dp}_k^F (1 + h_{2k}) + \text{dp}_k^C, & \text{dp}_{k+1}^C &= (\text{dp}_k^F + \text{dp}_k^C) h_{2k+1}. \end{aligned}$$

Then:

(a) (**Counting**). *Setting all $h_j = 1$ the recursion becomes*

$$\text{cnt}_{k+1}^F = 2 \text{cnt}_k^F + \text{cnt}_k^C, \quad \text{cnt}_{k+1}^C = \text{cnt}_k^F + \text{cnt}_k^C,$$

and after p steps $\text{cnt}_p^F + \text{cnt}_p^C = F_{m+2}$, recovering $|\mathcal{Z}_n| = F_{m+2}$ in $O(m)$ time.

(b) (**Product-sum**). *For the product-formula approximation*

$$H_{\text{approx}}(\mathbf{z}) = \prod_{k: z_k=1} h_k,$$

after p steps,

$$\sum_{\mathbf{z} \in \mathcal{Z}_n} H_{\text{approx}}(\mathbf{z}) = \text{dp}_p^F + \text{dp}_p^C.$$

Both computations run in $O(m)$ arithmetic operations, versus $O(F_{m+2}) = O(\varphi^m)$ by naive enumeration.

Proof. The pair automaton has states F (“free”: last emitted bit was 0, or start of string) and C (“constrained”: last emitted bit was 1). At each pair step k the valid emissions are:

$$\begin{array}{ll} F \xrightarrow{00} F & (\text{neither tile active}) \\ F \xrightarrow{10} F & (\text{left tile active, } h \text{ multiplied by } h_{2k}) \\ F \xrightarrow{01} C & (\text{right tile active, } h \text{ multiplied by } h_{2k+1}) \\ C \xrightarrow{00} F & (\text{neither tile active}) \\ C \xrightarrow{01} C & (\text{right tile active, } h \text{ multiplied by } h_{2k+1}) \end{array}$$

and the transition $C \xrightarrow{10}$ is *forbidden* (it would place a 1 immediately after the 1 that moved us to state C , creating consecutive 1s at the pair boundary).

Tracking $\text{dp}_k^F = \sum_{\mathbf{z} \text{ in state } F \text{ after } k \text{ pairs}} H_{\text{approx}}(\mathbf{z})$ and similarly for dp_k^C , the contributions of the three F -transitions and two C -transitions give exactly the stated recursion.

At $k = 0$ there is one tiling in state F (the empty string, $H_{\text{approx}} = 1$) and none in C , giving the initial conditions. Summing over all terminal states gives the total product-sum.

For part (a), set $h_j = 1$: each “ h multiplication” becomes $\times 1$, so dp_k^F counts tilings in state F and dp_k^C counts those in C . The state F can be reached via 00 from F (1 way), 10 from F (1 way), and 00 from C (1 way): total weight $2\text{cnt}_k^F + \text{cnt}_k^C$. State C via 01 from either: $\text{cnt}_k^F + \text{cnt}_k^C$. This gives the transfer matrix $M' = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, whose eigenvalues are φ^2 and $1/\varphi^2$ and which satisfies M'^p giving F_{2p+2} and F_{2p+1} in the appropriate entries, consistent with the pair FSM Theorem (Theorem ??). $\square$

TABLE 11. Pair DP speedup: $O(m)$ vs $O(F_{m+2}) = O(\varphi^m)$ naive enumeration. Verified at $n = 5$ ($m = 6, 21$ tilings) and $n = 6$ ($m = 10, 144$ tilings).

m	F_{m+2}	Speedup	m	Speedup
6	21	$7\times$	25	$16\,368\times$
10	144	$29\times$	30	$145\,221\times$
15	1 597	$228\times$	35	$1.4 \times 10^6\times$
20	17 711	$1\,771\times$	40	$1.3 \times 10^7\times$

Remark F.16 (Bijection $N \leftrightarrow T_N$). The Zeckendorf representation of any $N \in \{0, \dots, F_{m+2} - 1\}$ uses Fibonacci numbers $F_{a_1} > F_{a_2} > \dots > F_{a_r}$ (standard indexing, $F_1 = F_2 = 1$) with no two consecutive; the corresponding tiling T_N has active tile $a_i - 1$ (0-indexed) for each i . This gives an explicit $O(m)$ bijection $N \leftrightarrow T_N$ which is also computable by the pair automaton’s *address decoder*: the automaton state at pair k encodes which tilings have $N < F_{2k+2}$ (state F) vs $N \in [F_{2k+2}, F_{2k+3})$ (state C).

Theorem F.17 (Product formula for disjoint-window tilings). *Let $\mathbf{z} \in \mathcal{Z}_n$ be a Zeckendorf tiling with active tiles $\mathcal{A} = \{(b_{i_1}, a_{i_1}), \dots, (b_{i_r}, a_{i_r})\}$. Say the windows are **disjoint** if the half-open intervals $[b_{i_j}, a_{i_j})$ are pairwise disjoint. If the windows are disjoint,*

$$H(T_{\mathbf{z}}) = \prod_{j=1}^r (1 + 2^{r_{i_j}-1}).$$

Proof sketch. When the range windows $[b, a)$ are disjoint, the backward arcs occupy vertex-disjoint segments of the base Hamiltonian path. Odd cycles from distinct tiles share no vertices, so the conflict graph $\Omega(T_{\mathbf{z}})$ decomposes as a disjoint union $\Omega_1 \sqcup \dots \sqcup \Omega_r$. The independence polynomial factorises: $I(\Omega, 2) = \prod I(\Omega_j, 2)$. By the atomic formula (Theorem ??), $I(\Omega_j, 2) = H(\text{single-tile range } r_j) = 1 + 2^{r_j-1}$. $\square$

Remark F.18 (Chain correction for adjacent windows). When two tiles (b_1, a_1) and (b_2, a_2) are **chained** ($a_1 = b_2$, i.e., the windows share an endpoint), the conflict graph has additional cycle edges across the two segments. Computation at $n = 5$ shows the correction:

$$\alpha_1^{\text{true}} - \alpha_1^{\text{product}} = 2^{r_{\text{chain}}-3},$$

where $r_{\text{chain}} = a_2 - b_1$ is the combined range. Thus

$$H(T_{\mathbf{z}}) = \prod_{j=1}^r (1 + 2^{r_j-1}) - 4 \cdot 2^{r_{\text{chain}}-3} = \prod_{j=1}^r (1 + 2^{r_j-1}) - 2^{r_{\text{chain}}-1}$$

for a single chained pair of tiles. The general multi-chain correction formula is open.

APPENDIX G. GRID-SYMMETRY, SELF-COMPLEMENTARITY, AND THE FOUR-CATEGORY DECOMPOSITION

The tiling model carries an involution that precisely separates SC from NS tournament classes. We prove this and extract a complete four-category decomposition verified through $n = 7$ ($2^{15} = 32768$ tilings, 456 iso classes).

G.1. The TRANS_MAP involution. Fix n and let $\mathcal{T} = \{0, 1\}^m$ be the tiling space ($m = \binom{n-1}{2}$). Recall tiles (x, y) with $x > y + 1$, $1 \leq y < x \leq n$, ordered as in the explorer. Define the **TRANS_MAP** involution $\tau : \mathcal{T} \rightarrow \mathcal{T}$ by

$$\tau(M) \text{ has bit at tile } (x, y) = \text{bit of } M \text{ at tile } (n - y + 1, n - x + 1).$$

A tiling M is **grid-symmetric** iff $M = \tau(M)$. The count of grid-symmetric tilings is 2^{f_n} where $f_n = s + (m - s)/2$ (s = number of self-paired tiles $x + y = n + 1$).

Let $\sigma_n \in S_n$ be the involution $\sigma_n(v) = n + 1 - v$.

Lemma G.1 (TRANS_MAP = complement + relabeling). *For every tiling M : $T(\tau(M)) = (T(M)^{\text{op}})^{\sigma_n}$, where T^{op} denotes the full arc-complement.*

Proof. Tile (x, y) in M has bit = 0 iff $x \rightarrow y$ in $T(M)$. Under $T(M)^{\text{op}}$ this arc is reversed; applying σ_n relabels $v \mapsto n + 1 - v$, placing the arc at tile $(n + 1 - y, n + 1 - x) = (n - y + 1, n - x + 1)$. This matches $\tau(M)$'s bit at $(n - y + 1, n - x + 1)$. Base-path arcs are verified separately: σ_n sends $k \rightarrow k - 1$ to $(n + 1 - k) \rightarrow (n + 2 - k)$, and complementing restores the base-path direction. $\square$

G.2. Main theorem.

Theorem G.2 (Grid-symmetry $\Leftrightarrow$ self-complementarity).

- (i) *A tiling M is grid-symmetric iff $T(M)$ is self-complementary: $[T(M)] = [T(M)^{\text{op}}]$.*
- (ii) (Surjectivity) *Every SC iso class contains at least one grid-symmetric tiling.*
- (iii) (Odd blue count) *For every SC class C , the blue count $\text{bl}(C) := |\{M \in C : \tau(M) = M\}|$ is a positive odd integer.*
- (iv) *Every NS class has $\text{bl} = 0$.*

Proof. (iv). NS means $[T^{\text{op}}] \neq [T]$. By Lemma G.1, τ maps C to $[(T^{\text{op}})^{\sigma_n}] = [T^{\text{op}}] \neq C$. A grid-symmetric tiling $M = \tau(M) \in C$ would also lie in $[T^{\text{op}}]$, contradiction.

(i), direction $\Rightarrow$. $\tau(M) = M$ implies $T(M) = (T(M)^{\text{op}})^{\sigma_n}$, so $T(M)^{\sigma_n} = T(M)^{\text{op}}$, hence $T(M)^{\text{op}} \cong T(M)$.

(ii) and (iii). SC means τ maps C to itself (from (iv)'s contrapositive). So $\tau|_C$ is an involution on C . By the Parity Theorem, $|C| = H/|\text{Aut}|$ is odd. An involution on a set of odd cardinality has an odd positive number of fixed points. $\square$

Corollary G.3 (Exact characterization of grid-symmetry). *A tiling M is grid-symmetric if and only if the labeled tournament $T(M)$ satisfies*

$$T(M)^{\sigma_n} = T(M)^{\text{op}}$$

as labeled tournaments, where $\sigma_n(v) = n + 1 - v$. In other words, σ_n is the complementation permutation of $T(M)$. Verified computationally for $n = 4, 5, 6$ (perfect agreement across all 2^m tilings).

Remark G.4 (Formula for the blue count). For a SC class C with representative T_0 and complementation permutation σ_0 (so $T_0^{\sigma_0} = T_0^{\text{op}}$):

$$\text{bl}(C) = |\{\text{tilings } M \in C : T(M)^{\sigma_n} = T(M)^{\text{op}}\}| = |\{\tau \in S_n : T_0^\tau \text{ is a valid tiling and } \tau^{-1}\sigma_0\tau = \sigma_n\}|/|\text{Aut}(C)|$$

For $|\text{Aut}(C)| = 1$: $\text{bl}(C)$ counts the conjugators from σ_0 to σ_n whose image T_0^τ lands in the tiling space. Since σ_n and σ_0 must be conjugate in S_n (same cycle type) for any solution to exist, the formula gives $\text{bl}(C) = 0$ when cycle types differ and the class is NS.

G.3. Four-category decomposition.

Corollary G.5. *Every merged node of $G_n/\mathbb{Z}_2$ is exactly one of:*

- **Pure blue** (SC, $\text{bl} = |C|$, $\text{bk} = 0$): all tilings grid-symmetric; τ acts as identity on C .
- **Mixed** (SC, $0 < \text{bl} < |C|$, $\text{bk} > 0$): both grid-sym and non-grid-sym tilings; τ has nontrivial orbits on C .
- **Pure black** (NS merged pair, $\text{bl} = 0$, $\text{bk} = |C_1| + |C_2|$).
- **Zero** ($\text{bl} = \text{bk} = 0$): impossible by Theorem G.2(ii).

The global constraints are: $\sum_{\text{SC}} \text{bl}(C) = 2^{f_n}$ and $\sum_{\text{SC}, \text{mix}} \text{bk}(C) + \sum_{\text{NS-pr}} 2|C_i| = 2^m - 2^{f_n}$.

n	m	2^m	2^{f_n}	#SC	#NS-pr	#mix	#pb	#pblk
3	1	2	2	2	0	0	2	0
4	3	8	4	2	1	1	1	1
5	6	64	16	8	2	5	3	2
6	10	1024	64	12	22	10	2	22
7	15	32768	512	88	184	84	4	184

TABLE 12. Four-category decomposition for $n = 3-7$. #pb=pure blue, #pblk=pure black NS-pairs. Verified by exhaustive canonicalization ($n = 7$: ≈ 70 s numpy computation).

G.4. Data through $n = 7$.

Blue value multiplicities. The count of SC classes with each blue value at $n = 3-7$:

n	bl=1	bl=3	bl=5	bl=7	bl=9	#SC
3	2	0	0	0	0	2
4	1	1	0	0	0	2
5	4	4	0	0	0	8
6	2	2	2	4	2	12
7	10	12	16	32	18	88

The count-vector $(c_1, c_3, c_5, c_7, c_9)$ satisfies $\sum c_{2k-1} = \#SC$ and $\sum (2k-1)c_{2k-1} = 2^{f_n}$, two constraints on five unknowns. At $n = 6$ and $n = 7$ the value $bl = 7$ has the highest count ($\approx 33\text{--}36\%$ of SC classes), a pattern deserving explanation.

Persistent pure-blue classes. A class is pure blue iff τ acts as the identity on all its tilings. The following classes appear consistently:

H	$ \text{Aut} $	present at n
1	1	all $n \geq 3$ (transitive tournament, $\#\text{tilings} = 1$)
3	3	odd $n \geq 3$ ($n = 3, 5, 7, \dots$)
9	9	$n \geq 6$
15	5	odd $n \geq 5$ ($n = 5, 7, \dots$, $\#\text{tilings} = 3$)

Each has $\#\text{tilings} = H/|\text{Aut}| \leq 3$, so the pure-blue condition is forced: parity leaves no room for non-fixed tilings.

NS-pair structure. For NS-pair (C, C^{op}) : $H(C) = H(C^{\text{op}})$ (Hamiltonian paths biject via reversal) and $|\text{Aut}(C)| = |\text{Aut}(C^{\text{op}})|$, so both classes have the same odd tiling count. Merged black count $= 2H/|\text{Aut}|$, always even. At $n = 7$: 184 NS-pairs, $\sum bk = 25460$.

MULTI-AGENT MATHEMATICS RESEARCH PROJECT

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